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Engineering Functional CLA-Targeting CAR Approaches for Pancreatic Ductal Adenocarcinoma

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Abstract

Pancreatic cancer remains a highly lethal malignancy with limited therapeutic options. Chimeric antigen receptor (CAR) therapy has revolutionized the treatment of hematological cancers but still faces major limitations in solid tumors, particularly due to the scarcity of tumor-specific targets. Cutaneous lymphocyte antigen (CLA) recently emerged as a promising PDAC target due to its high tumor expression and limited presence in healthy tissues. However, previously reported CLA-directed CAR constructs lacked antitumor functionality. Here, we investigated multiple strategies to generate functional CLA-targeting CAR approaches. We first hypothesized that impaired activity resulted from fratricide caused by CLA expression on activated T cells. CLA knockout was successfully achieved through deletion of fucosyltransferase-7, but not by knockout of the major CLA carrier backbones CD162, CD44 or CD43, suggesting additional CLA carriers or compensatory regulation. As CLA knockout alone did not restore CAR-mediated killing, we explored whether insufficient binding affinity limited CAR activity. Affinity maturation was performed *in silico* and *in vitro* using yeast surface display, identifying 39 candidate mutations, although none restored cytotoxicity. We finally switched to an AdCAR strategy using anti-biotin CAR T cells combined with biotinylated anti-CLA scFv-Fc adapters. This approach enabled efficient, concentration-dependent cytotoxicity with both CLA-targeting binders. Additionally, we identified a dynamic, cell density-dependent regulation of CLA expression. Finally, glycan profiling of CLA binders further revealed broader-than-expected glycan interactions, suggesting a potentially wider definition of the CLA family. Overall, our findings establish CLA as a functional PDAC immunotherapy target while revealing unexpected complexity in its regulation and molecular presentation.

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1 Introduction

Pancreatic ductal adenocarcinoma (PDAC) is a highly lethal disease, currently the third leading cause of cancer-related death in the U.S. and Europe, accounting for ~8% of deaths despite only ~3% of cancer cases[1,2]. Therapeutic options remain limited to chemoradiotherapy and surgery, yet ~80% of patients present with unresectable disease, resulting in a poor ~13% 5-year survival rate[3,4]. This underscores the urgent need for novel therapeutic strategies[5]. Chimeric antigen receptor (CAR) T cells have transformed adoptive immunotherapy, becoming standard-of-care for multiple hematological malignancies[6]. These genetically engineered cells specifically recognize and eliminate target cells[7]. However, their efficacy in solid tumors remains limited, largely due to the challenge of identifying tumor-restricted antigens[8]. This highlights the importance of target selection for safe and effective CAR T cell therapy in solid tumors.

Recently, four promising PDAC-associated targets were identified: CD318, TSPAN8, CD66c and cutaneous lymphocyte antigen (CLA)[9]. CARs targeting each of these four targets have been generated, with CD318 advancing to a phase I/IIa clinical trial (NCT07153289). CLA is particularly attractive due to its favorable safety profile and restricted expression on a subset of peripheral leukocytes. Beyond PDAC, CLA is also overexpressed in lymphoma[10]. However, CLA-directed CARs have not yet demonstrated functional activity, warranting further investigation[9].

CLA was described as a carbohydrate epitope expressed on circulating leukocytes that mediates skin homing[11] and is defined by binding of the HECA-452 IgM antibody[12]. Although primarily associated with memory T cells, CLA is also expressed on memory B cells, neutrophils and innate lymphoid cells[13]. Depletion of specific immune subsets can be clinically managed, as demonstrated by CD19-targeting CAR T cell therapies[14], making CLA still a promising target candidate. CLA biosynthesis depends on glycosylation mediated by glycosyltransferases, particularly fucosyltransferase VII (FUT7)[15]. Initially described on P-selectin glycoprotein ligand-1 (CD162/PSGL1/SELPLG)[11], CLA has also been identified on CD44[16] and CD43[17]. Structurally, CLA consists of sialyl Lewis X-related motifs with additional fucosylation and, in some reports, sulfation[18,19], although its precise molecular definition remains incomplete. Functionally, CLA binds E- and P-selectin on dermal endothelial cells, enabling leukocyte tethering and rolling during extravasation, thereby promoting recruitment to inflamed skin[20]. Elevated CLA^+ T cells frequencies are observed in psoriasis, atopic dermatitis and cutaneous T cell lymphoma[21]. The combination of CLA overexpression in PDAC, the availability of a well-characterized binding antibody, and its restricted expression on a dispensable subset of leukocytes collectively make CLA an attractive target for CAR T cell therapy. However, important biological aspects of CLA remain poorly understood, potentially limiting the development of effective CLA-directed cellular therapies.

In this study, we aimed to address the limitations preventing the development of functional CLA-directed CAR T cells. We identified a density-dependent regulation of CLA expression and investigated the underlying mechanisms. In parallel, we explored the structure and cellular localization of CLA, which remain incompletely defined and inconsistently described in literature.

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2 Methods

Cell lines and culture conditions

AsPC1, BxPC3, HL60, SupT1, HEK293T, PanCa0203 and Capan2 cells (ATCC) were cultured in RPMI or DMEM high-glucose medium (Biowest) supplemented with 2 mM L-glutamine and 10% fetal bovine serum (FBS; Eximus). OCI-AML2 (DSMZ) cells were cultured in Alpha-MEM Eagle medium (PAN-Biotech) with 20% FBS. PanCa0201 primary PDAC cell line was generated from dissociated human PDAC biopsies using the Tumor Dissociation Kit, gentleMACS™ Octo Dissociator with Heaters, and Tumor Cell Isolation Kit, followed by culture in Pancreas TumorMACS™ medium (Miltenyi Biotec). *S. cerevisiae* EBY100 (ATCC) were cultured in YPD, SDCAA or SGCAA media, as previously described[22]. For cell-density experiments, cells were seeded at defined densities and cultured under indicated conditions, including variable FBS concentrations, stress conditions, or supplementation with IL-12, IL-4 or TGF-β (Miltenyi Biotec). Marker expression was daily analyzed by flow cytometry.

Flow cytometry

Samples were acquired on MACSQuant® Analyzer 10 or 16 instruments (Miltenyi Biotec) and analyzed using MACSQuantify™ (Miltenyi Biotec, v2.13.0), FlowJo (BD Biosciences, v10.10.0), GraphPad Prism (GraphPad Software, v10.1.2) and Microsoft Excel (Microsoft, v2502). Staining with Miltenyi Biotec antibody conjugates was performed according to manufacturer instructions. Dead cells were excluded using 7-AAD. All antibodies used in this study are listed in supplementary table 1. Gating strategies are shown in supplementary figure S1.

Cyclic immunofluorescence

Primary human PDAC tissues were collected at the University Medical Center Göttingen from patients undergoing surgical resection of tumor masses. Fresh-frozen healthy organ tissue samples were purchased from ProteoGenex and BioIVT. Tissues were prepared, fixed and stained as previously described[9]. Imaging was performed on the MACSima™ Imaging Platform (Miltenyi Biotec) according to the manufacturer's instructions. For figure display, all HDR images were uploaded in ImageJ (v1.49) and the histogram was set to equal values for all images from 0 to 29980.

CRISPR/Cas9 knock-out (KO) generation

CD162, CD44, CD43 and FUT7 KO were generated by CRISPR-Cas9 electroporation using the CliniMACS® Electroporator integrated into the CliniMACS® Prodigy system (Miltenyi Biotec). The three recommended predesigned sgRNAs (AA, AB and AC), Cas9 Nuclease V3 and electroporation enhancer were obtained from IDT and used according to manufacturer recommendations. The KO in T cells was performed three days after isolation.

CAR/AdCAR construct generation and lentiviral production

Synthetic gBlocks (IDT) were cloned into Miltenyi Biotec backbones containing LacZ and kanamycin resistance cassettes. Bve1 restriction enzymes, T4 ligase and cloning reagents were provided by NEB, according to their standard protocol. Plasmids were transformed into 5-alpha competent *E-coli* (NEB) and purified using Zyppy™ Miniprep and ZymoPURE™ Maxiprep kits (Zymo research). AdCAR constructs were generated as previously described[23]. Lentiviral (LV)

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particles were produced by transient polyethylenimine-mediated transfection of HEK293T cells. Viral supernatants were collected after 48 h, concentrated by overnight centrifugation, resuspended in TexMACS™ medium (Miltenyi Biotec) and stored at −70°C. Viral titers were determined by SupT1 transduction.

T cell isolation and CAR/AdCAR transduction

Peripheral blood mononuclear cells (PBMCs) were isolated by PanColl (PanBiotech) density-gradient centrifugation from whole blood. T cells were purified using the Pan T Cell Isolation Kit and activated in TexMACS™ medium containing 1:100 TransAct™, IL-7 and IL-15 (12.5 ng/mL each; all Miltenyi Biotec). Cells were transduced with lentivirus 24h after activation. Three days later, TransAct™ was removed and cells were cultured in IL-7/IL-15-supplemented medium. Transduction efficiency was assessed by flow cytometry.

Anti-CLA scFv-Fc production

Anti-CLA scFv-Fc expression constructs were synthesized, cloned into the constitutive mammalian expression vector pcDNA™3.4 TOPO™ and prepared by GeneArt (ThermoFisher Scientific). Expi293F cells were expanded and transiently transfected according to the manufacturer's instructions using the ExpiFectamine™ 293 Transfection Kit (ThermoFisher Scientific). Four days post-transfection, cultures were clarified by addition of 100 g/L Celite 545 filter aid (Sigma-Aldrich) followed by sterile filtration through a 0.22 µm PES membrane (Nalgene™ Rapid-Flow™, ThermoFisher Scientific). Cell-free supernatants were stored at -20°C until purification. ScFv-Fc proteins were purified using a HiTrap™ MabSelect column (Cytiva) according to the manufacturer's protocol and buffer-exchanged into PBS (pH 7.4) using PD-10 Sephadex™ G-25 desalting columns (Cytiva). Protein purity was assessed by SDS-PAGE under reducing conditions. Samples were mixed with 5× SDS sample buffer containing DTT, heated at 95°C for 5 min, and separated on 4-20% Criterion™ TGX™ Precast Midi Protein Gels (Bio-Rad). Electrophoresis was performed at 220 V for 50 min, followed by Coomassie Blue staining for 30 min at room temperature.

Adapter biotinylation

Pure antibodies (Miltenyi Biotec) and scFv-Fc molecules were buffer-exchanged into 0.1 M NaHCO3 solution (Sigma-Aldrich) via NAP-5 columns (Cytiva). Proteins were biotinylated for 1h at room temperature using a 2-fold molar excess of biotin-LC-LC-NHS dissolved in anhydrous DMSO (both ThermoFisher Scientific). Samples were purified on NAP-5 columns, rebuffered in PBS (VWR) and quantified by absorption at 280 nm using a Nanodrop spectrophotometer (ThermoFisher Scientific). Successful biotinylation was validated by flow cytometry using anti-biotin-PE staining.

Cytotoxicity assays and live-cell imaging

Target cells (2×10^4/well) were co-cultured with CAR^+ or $AdCAR^+$ T cells at indicated effector-to-target (E:T) ratios. For AdCAR assays, biotinylated adapters were added at specified concentrations. Cytotoxicity was monitored as reduction in GFP-positive area using the IncuCyte® S3 Live-Cell Analysis System (Sartorius) and the supplied software (v2019A and v2023A). GFP area was normalized to the first time point. Following co-culture, T cell activation markers were analyzed by flow cytometry and cytokines quantified from supernatants using the MACSPlex™

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Cytokine 12 Kit (Miltenyi Biotec). Data were analyzed with MACSPlexInspectoR software (https://www.miltenyibiotec.com/DE-en/resources/tools/macsplex-inspector.html).

Immunoprecipitation, SDS-PAGE and peptide mass fingerprinting

For each condition, 1×10^7 cells were lysed for 30 min on ice in Ecosurf™ EH-9 lysis buffer (Miltenyi Biotec) containing Halt™ Protease Inhibitor Cocktail (ThermoFisher Scientific). Immunoprecipitation was performed using antibodies with Protein A or biotinylated antibodies with anti-biotin microbeads, followed by µMACS™ Protein A/G separation (all Miltenyi Biotec), according to the manufacturer recommendations. Immunoprecipitates were reduced with 10 mM DTT for 5 min at 95°C and alkylated with 20 mM iodoacetamide (IAA) for 30 min at room temperature. Samples were separated on Novex™ WedgeWell™ 4-20% Tris-Glycine gels using the XCell SureLock™ Mini-Cell system with Tris-Glycine SDS running buffer (all ThermoFisher Scientific). Gels were stained with Bio-Safe™ Coomassie G-250 (Bio-Rad) for 1h and destained overnight under agitation. Bands of interest were excised, destained and digested overnight with trypsin (20 µg/mL) in 50 mM NH4HCO3. Peptides were analyzed by peptide mass fingerprinting (PMF) using an Orbitrap Fusion Lumos Tribrid Mass Spectrometer (ThermoFisher Scientific). Proteins were identified by database searching against the UniProt database.

Affinity maturation

In silico affinity maturation was performed using the Efficient Evolution of Human Antibodies from Protein Language Models package (v0.1)[24]. For yeast surface display (YSD), WT and mutated CLA scFv OligoPools (IDT) were cloned into pYSDM2.0 by yeast recombinational cloning, as previously described[25]. Libraries containing 1-6 random mutations per clone, restricted to the CDR regions, were transformed into electrocompetent *S. cerevisiae* EBY100 as previously described[26], resulting in libraries in which each yeast clone displays a single scFv variant. Following induction in SGCAA medium[22], scFvs were labelled via HA-tag APC staining and non-expressing clones depleted using anti-APC microbeads and LD/LS columns (Miltenyi Biotec). Negative selection was performed by incubating yeast with CLA-deficient HL60 FUT7 KO cells at a 1:10 target-to-yeast ratio, followed by sorting using a MACSQuant® Tyto® sorter (Miltenyi Biotec). Positive selection was conducted the following week under similar conditions using WT HL60 cells. To ensure complete CLA depletion in HL60 FUT7 KO cells and robust CLA expression in HL60 WT cells, cells were respectively depleted or enriched using anti-CLA antibodies and LD/LS columns (Miltenyi Biotec) immediately prior to co-culture. After selection, single clones were isolated on agar plates, induced, and screened in 96-well format against WT or FUT7 KO HL60 cells. Clones were analyzed by flow cytometry and ranked according to WT/KO binding ratios and sequenced (Eurofins) to identify mutations.

Multiplex RT-qPCR

HL60, OCI-AML2 and T cells were cultured at varying densities for 5 days before RNA extraction using the RNeasy® Mini Kit (Qiagen). RNA integrity and concentration were assessed using Qubit™ RNA HS kit (ThermoFisher Scientific) and RNA ScreenTape 4150 TapeStation assay (Agilent Technologies). Multiplex RT-qPCR was performed using the QuantiTect Multiplex RT-PCR NoROX Kit (Qiagen) on a CFX96 instrument (Bio-Rad) with 50 ng RNA per reaction. Expressions of FUT7, B4GALT1, GCNT1, FUT4, ST3GAL4 and B3GNT3 were quantified using IDT primer/probe mixes with HPRT1 as housekeeping gene. Thermal cycling conditions consisted

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of reverse transcription at 50°C for 20 min, initial activation at 95°C for 15 min, followed by 50 cycles of 94°C for 45s and 60°C for 75s. Relative gene expression was calculated by the ΔΔCt method, where ΔCt = Ct(gene of interest) − Ct(HPRT1) and ΔΔCt = ΔCt(sample) − ΔCt(control). Fold changes in gene expression were calculated as 2^(−ΔΔCt).

Glycan array screening

A 300-glycan array was performed by CD BioGlyco using HECA-452 and REA1101 CLA antibodies together with generated CLA scFv-Fc binders. CD66c antibody served as background control. Samples were tested at 0.03 mg/mL. Positive control signal intensities were used for data normalization after background subtraction. After normalization and background subtraction performed by CD BioGlyco, log10 fold changes relative to CD66c controls were calculated and z-score normalized. Glycans with z-scores >3 were selected. Raw data and detailed analyses can be downloaded in the supplementary materials.

3 Results

CLA is highly expressed in PDAC but previously reported direct CLA CARs lack functionality

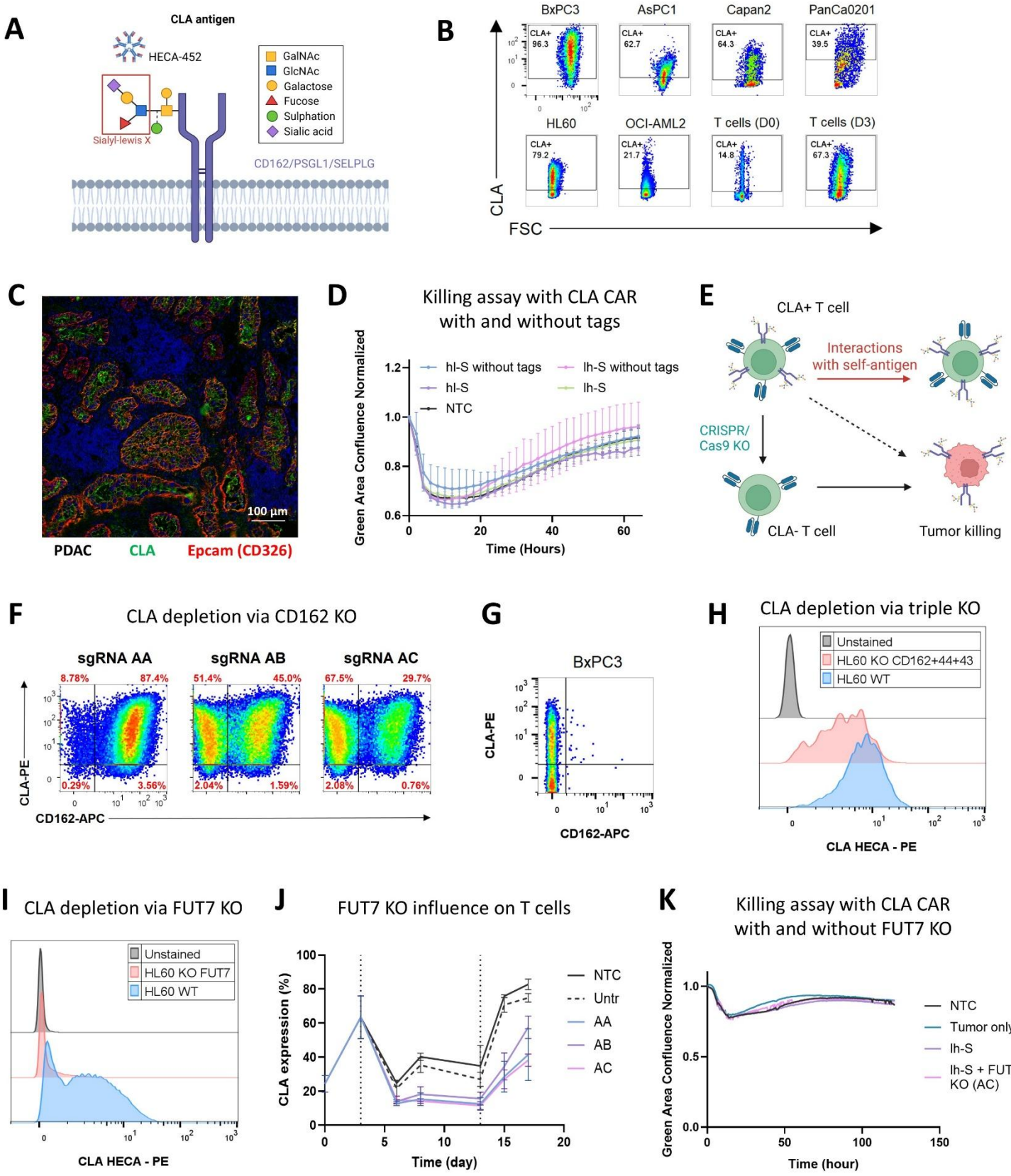
CLA has been described as a glycosylated sialyl-Lewis X epitope mainly carried on CD162/PSGL-1 (figure 1.A). It is highly expressed on multiple PDAC cell lines (BxPC3, AsPC1, Capan2, PanCa0201) and on patient tissues (figures 1.B-C). CLA is also detected on leukemia cell lines (HL60, OCI-AML2), suggesting broader relevance (figure 1.B). On T cells, basal CLA expression increases upon activation (figures 1.B, S2), consistent with prior reports[27]. Healthy tissue array analysis showed no or only minor background CLA expression outside PDAC (figure S3), supporting a favorable safety profile. The only exception was the kidney cortex, where the CLA antibody appeared to bind to an undefined intercellular region. This staining may reflect secreted antigen binding or nonspecific antibody reactivity and warrants further investigation. However, previously reported scFv based anti-CLA CARs were non-functional for unclear reasons[9]. CLA has a prominent expression on activated T cells and previous constructs missed tags for identification of successful surface expression of the CAR, leaving it unclear whether the dysfunction was due to a lack of expression or other reasons. Myc/His tags can help with CAR identification but can interfere with activity and folding. We therefore generated tagged and untagged constructs (figure S4.A), but all failed to induce killing in two PDAC models (figures 1.D, S4.B).

FUT7-mediated CLA depletion does not restore CAR function

Because activated T cells express CLA, we hypothesized that reduced CAR activity was due to cis- or trans-interactions with the self-antigen (figure 1.E)[28]. Since CLA is a glycosylated epitope, direct KO is challenging. We first targeted CD162, the main proposed backbone[11]. Although two sgRNAs (AB, AC) efficiently reduced CD162, CLA levels surprisingly remained unchanged, indicating additional carriers (figure 1.F). Consistently, BxPC3 cells strongly expressed CLA despite lacking CD162 (figure 1.G). As CD44 and CD43 have also been reported as CLA carriers[29], we generated CD162/CD44/CD43 triple KO cells, but CLA decreased only marginally (10% in HL60 and 4% in activated T cells; figures 1.H, S5). Immunoprecipitation followed by SDS-PAGE and PMF analysis failed to identify alternative backbones, likely due to nonspecific antibody binding (figure S6).

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Figure 1



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Figure 1. CLA expression and functionality of CLA CAR constructs following different CLA knockout strategies.

A) *Schematic of the CLA antigen, a glycosylated epitope containing sialyl-Lewis X, previously described on CD162/PSGL1/SELPLG and recognized by HECA-452 IgM. Created with BioRender.com.* ***B)*** *Flow cytometry analysis of CLA expressions in PDAC cell lines (BxPC3, AsPC1, Capan2, PanCa0201), leukemia cell lines (HL60, OCI-AML2) and T cells at day 0 post-isolation (D0) and day 3 post-activation (D3). Representative of a technical duplicate shown. Gating strategy and controls are provided in figure S1.B.* ***C)*** *Cyclic immunofluorescence (IF) of representative human PDAC tissue showing co-expression of CLA and cytokeratin-positive tumor cells. Image is representative of two cyclic IF runs from two different PDAC specimens.* ***D)*** *Killing curves showing GFP-based PanCa0201 tumor cell confluence, normalized to the first time point, for hl-S and lh-S CLA CAR constructs with or without Myc and His tags at an E:T ratio of 10:1, compared with NTC. Error bars represent the mean of two independent donors.* ***E)*** *Schematic of potential interactions with self-antigen caused by CLA expression on activated T cells and its prevention through CRISPR/Cas9-mediated KO. Created with BioRender.com.* ***F)*** *Flow cytometry analysis of CLA depletion in primary activated T cells following CD162 KO using sgRNAs AA, AB and AC.* ***G)*** *Flow cytometry analysis of CLA and CD162 expression in BxPC3 cells. Representative of a technical duplicate shown.* ***H)*** *Flow cytometry analysis of CLA depletion in HL60 cells following triple CD162/CD43/CD44 KO compared with WT. Gating strategy and controls are provided in figure S5.* ***I)*** *Flow cytometry analysis of CLA depletion in HL60 cells following FUT7 KO using sgRNA AC, compared with WT.* ***J)*** *CLA expression on peripheral T cells over time. Comparison of FUT7 KO efficiency using sgRNA AA, AB and AC, non-transduced electroporated control (NTC) and untreated control (Untr). Cells were activated with TransAct™ on days 0 and 13.* ***K)*** *Killing curves showing GFP-based PanCa0203 tumor cell confluence, normalized to the first time point, for lh-S CLA CAR construct with or without FUT7 KO (sgRNA AC), at an E:T ratio of 2:1, compared with NTC and tumor-only controls, from one donor.*

We therefore shifted to FUT7, a key enzyme in CLA biosynthesis. FUT7 KO completely abolished CLA and HECA IgM binding in HL60 cells (figure 1.I). In T cells, TransAct™ stimulation (days 0 and 13) strongly increased CLA, while FUT7 KO reduced this upregulation by ~50% versus controls (figure 1.J). Among the three sgRNAs tested (AA, AB and AC), sgRNA AC showed the highest efficiency and was selected for further experiments (figure S7.A). Interestingly, CLA regulation differed between subsets, with more sustained expression in $CD8^+$ cells (figure S7.B-C). In $CD4^+$ cells, CLA declined similarly in untreated and KO conditions, whereas $CD8^+$ cells showed a slower decrease, indicating more dynamic regulation in $CD4^+$ cells. Given that FUT7 may affect broader T cell biology, we further assessed the impact of KO. FUT7 KO did not significantly alter proliferation, activation marker expression (CD69, CD137, CD25, PD1) or differentiation into Tn, Tscm, Tcm, Tem and Teff subsets (figures S7.D-I). Despite efficient CLA depletion, combining FUT7 KO with CLA CARs (e.g. VLVH) did not restore tumor killing, indicating that self-interactions alone do not explain CAR inactivity (figure 1.K).

Affinity constraints may contribute to dysfunction, yet affinity maturation fails to rescue activity

We next hypothesized insufficient binding affinity. HECA-452 is an IgM antibody, which typically displays low intrinsic affinity but compensates through high avidity due to pentameric structure. CARs use monomeric scFv format, therefore reducing avidity, likely weakening target engagement (figure 2.A). A monomeric recombinant human IgG1 version (REA) confirmed reduced staining compared to IgM (~10-fold lower mean fluorescence intensity (MFI)), supporting reduced effective binding (figures S8.A-B).

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Figure 2

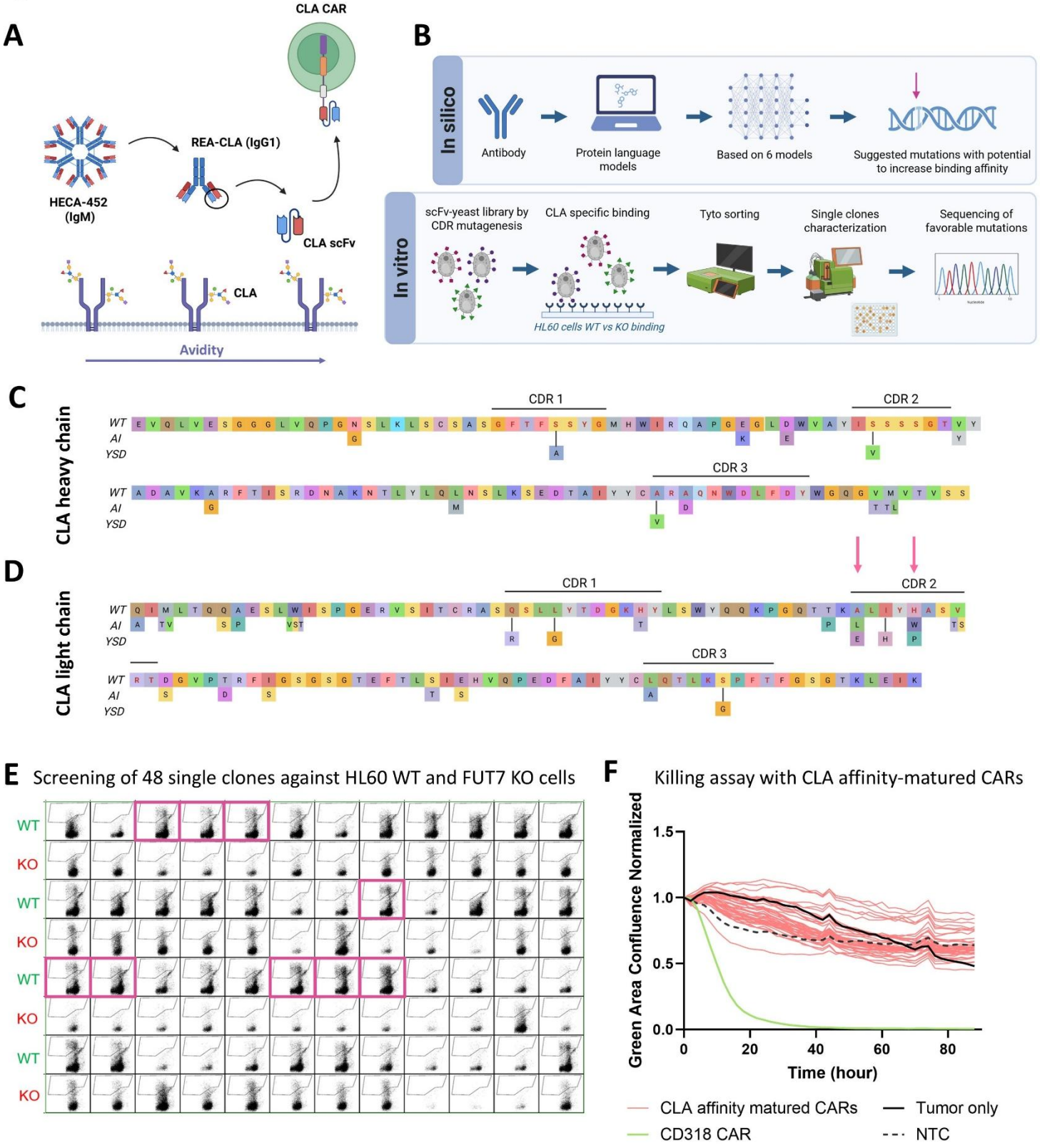


Figure 2. In vitro and in silico CLA affinity maturation to enhance target binding affinity.

A) *Schematic of potential affinity constraints contributing to CAR dysfunction. CLA scFv was derived from the REA IgG1, derived itself from the HECA-452 IgM, reducing binding valency and avidity, potentially impairing affinity.* ***B)*** *Affinity maturation workflow combining in silico protein language model prediction and in vitro yeast surface display (YSD).* ***C-D)*** *Summary of mutations identified by in silico (AI) and YSD approaches relative to the parental CLA heavy chain* ***C)*** *and light chain* ***D)*** *sequences. Two mutations in the*

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light-chain CDR2 region (indicated by arrows) were predicted by both methods. The CDRs were defined according to the Chothia numbering scheme. CDR, complementarity-determining region. ***A-D)*** *Created with BioRender.com.* ***E)*** *Flow cytometry screening of 48 single clones from a 96-well plate against HL60 WT and FUT7 KO cells. Clones within the pink frame were selected based on the ratio of % binding to WT versus KO cells.* ***F)*** *Killing curves showing GFP-based BxPC3 tumor cell confluence, normalized to the first time point, for 30 affinity-matured CLA CAR constructs at an E:T ratio of 2:1, compared with NTC, CD318 CAR and tumor-only controls. Data represents the mean of two independent donors.*

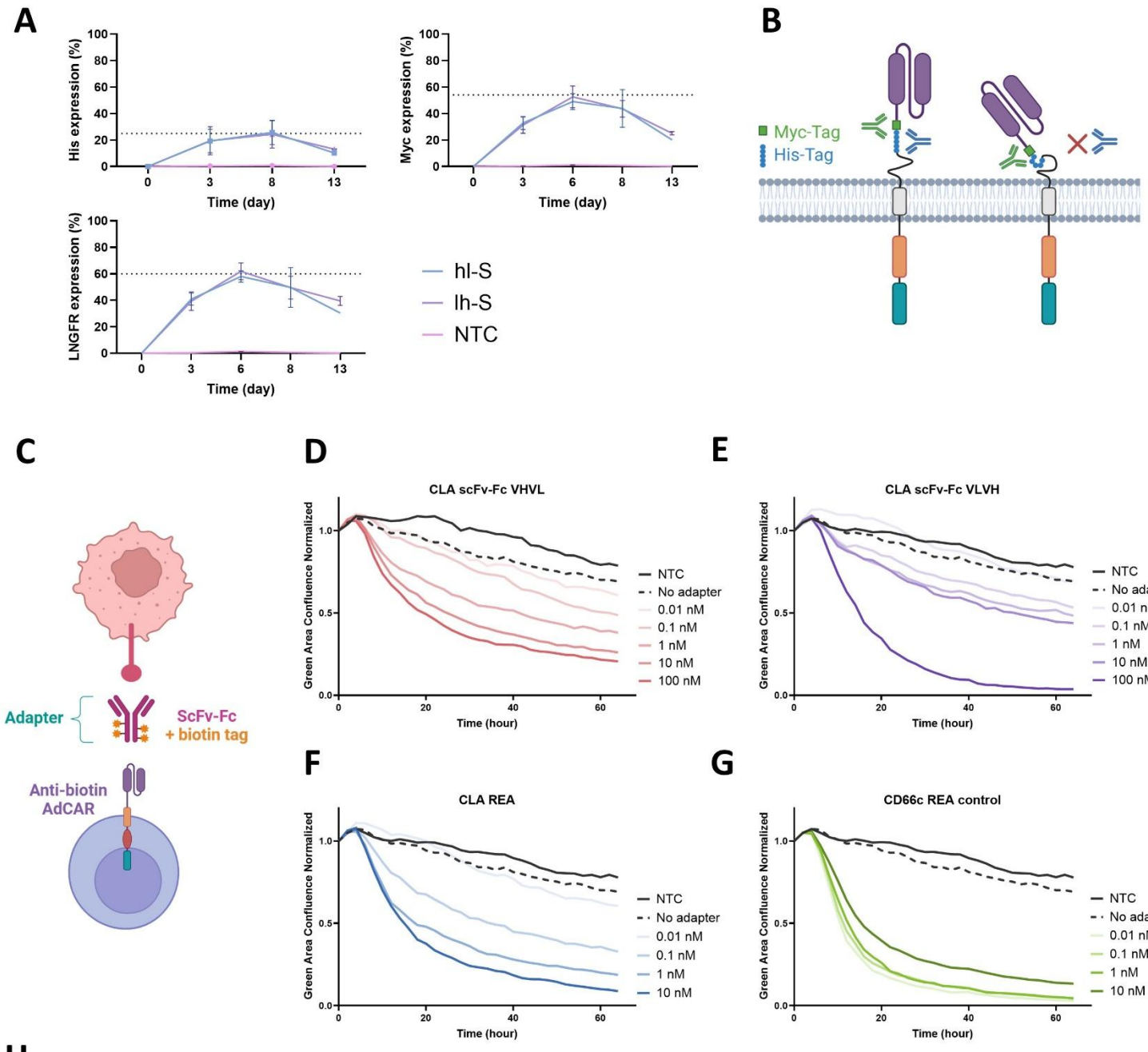
To improve binding to CLA, we pursued affinity maturation of the CAR's scFvs using complementary *in silico* and *in vitro* approaches (figure 2.B). First, we applied a protein language model-based affinity maturation workflow[24], which predicts evolutionarily plausible antibody mutations without structural or antigen-specific input. Using the anti-CLA scFv sequence, the model identified 30 candidate mutations across framework and CDR regions (figures 2.C-D), ranked by support from six integrated databases (supplementary table 2). In parallel, yeast surface display (YSD) was performed using HL60 cells instead of recombinant antigen (figure S9.A). Negative selection with FUT7 KO cells removed nonspecific binders, followed by positive selection on WT HL60 (figure S9.B). Sorting yielded 1.41% positive cells, which is within the expected range for such selections (figure S9.C). 96 single clones were expanded and re-screened on WT and FUT7 KO cells. Binding specificity was quantified based on the ratio of % binding to WT versus KO cells, enabling selection of target cell specific binders (figure 2.E). Twelve clones were selected showing ≥2.5-fold higher binding signal to WT cells based on frequency. Two clones were confirmed by sequencing, D10 and C2 (with a truncated heavy chain), containing six and three mutations, respectively (figures 2.C-D, S9.D; supplementary table 2). For functional validation, we selected *in silico* mutations supported by ≥4 models, together with YSD-derived mutation combinations (supplementary table 3). However, screening of the 40 selected candidates after affinity-maturation on BxPC3 cells failed to restore cytotoxicity (figure 2.F).

Functional CLA targeting is achieved via AdCAR using biotinylated scFv-Fc adapters

We next suspected CAR misfolding as a potential cause for CAR dysfunctionality. Indeed, analysis of CAR expression revealed discordant detection of the two encoded tags relative to the LNGFR transduction marker. Myc expression matched LNGFR (~60%), while His-tag was reduced (~24%), suggesting structural masking (figures 3.A-B). To overcome potential folding limitations which might result in affinity reduction, we switched to the AdCAR platform. In this system, biotinylated adapter molecules bind tumor cells and are recognized by anti-biotin AdCAR T cells (figure 3.C)[23]. We generated biotinylated CLA scFv-Fc adapters (VHVL/VLVH). Compared with the original CAR format, scFv-Fc adapters are likely to reduce misfolding risk and present similar avidity through Fc-mediated dimerization, resulting in two binding sites per molecule. Binding analysis on OCI-AML2 and BxPC3 cells showed similar affinities for VHVL and VLVH adapters, comparable to the validated CD66c REA control[30], and higher binding on BxPC3 cells. Both scFv-Fc adapters exhibited an approximately 100-fold lower EC50 value compared to CLA REA antibody, indicating a significantly higher apparent affinity (figure S10). Both adapters induced dose-dependent cytotoxicity of BxPC3 cells, with near-complete tumor clearance at high VLVH dose (figures 3.D-E). The CLA REA control also exhibited strong, concentration-dependent killing, with the highest activity observed at 10 nM adapter (figure 3.F). The CD66c positive control showed the strongest activity (figure 3.G). Results were reproduced in a second donor, showing similar overall trends despite less distinct dose separation (figure S11).

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Figure 3



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Figure 3. Functional characterization of CLA CARs using biotinylated CLA scFv-Fc adapter molecules in the AdCAR system.
A) *Comparison of CAR expression of hl-S and lh-S CLA CAR T cells over time, detected by His-tag, Myc-tag or LNGFR staining, compared with NTC. Error bars represent the mean of two independent donors.* ***B)*** *Schematic of tagged CLA CAR constructs at the cell surface. Left: correctly folded CAR with detectable Myc and His tags. Right: potential CAR misfolding or improper hinge configuration leading to masking of the His tag and loss of anti-His antibody detection. Created with BioRender.com.* ***C)*** *Schematic of the AdCAR system, which separates antigen recognition from CAR activation and induces T cell activation only in the presence of specific adapter molecules (e.g., biotinylated scFv-Fc). Created with BioRender.com.* ***D-G)*** *Killing curves showing GFP-based BxPC3 tumor cell confluence, normalized to the first time point, for CLA AdCAR constructs at an E:T ratio of 2:1 using a CLA scFv-Fc VHVL adapter* ***D)****, CLA scFv-Fc VLVH adapter* ***E)****, CLA REA adapter* ***F)****, or CD66c REA adapter* ***G)****, compared with non-transduced T cells (NTC) and AdCAR without adapter (no-adapter) controls, from one donor. Biotinylated adapters were tested at concentrations ranging from 0.01-100 nM for scFv-Fc adapters and from 0.01-10 nM for REA controls.* ***H)*** *Heatmaps summarizing CD69, 4-1BB, PD-1 and LAG-3 expression (%) as well as GM-CSF and IFN-γ secretion (pg/mL) across adapter conditions*.

Activation correlated with cytotoxicity: CD69, 4-1BB and PD-1 peaked at high VLVH/low CD66c doses, while LAG-3 was highest without adapter as CAR T cells remained inactive (figure 3.H). GM-CSF and IFN-γ production were strongest with CLA REA and CD66c, consistent with their potent killing activity. Interestingly, despite strong cytotoxicity, scFv-Fc adapters induced lower cytokine and activation levels than controls except at highest dose. Overall, CLA mediates effective CAR killing in AdCAR format, with limited fratricide impact.

CLA variability is density-dependent but remains mechanistically unresolved

We observed pronounced variability in CLA expression across days (figure 4.A). Variation was independent of FBS concentration and did not correlate with tested stress conditions, except for a CLA decrease under constant shaking (figures S12.A-C). Instead, a consistent inverse correlation with cell density was observed across HL60, T cells and OCI-AML2 (figures 4.B, S12.D-E). We next investigated the underlying regulatory factors driving this density-dependent downregulation.

We first assessed whether changes in known scaffold or carrier proteins could explain reduced CLA expression. Minor decreases in CD162 and CD44 were observed, but these were insufficient to account for the ~50% reduction in CLA at high density (figure 4.C), consistent with previous evidence that CLA is not restricted to these backbones. We next evaluated cytokine-mediated regulation, as increasing cell density alters the cytokine milieu. IL-12 has been reported to promote CLA via skin-homing programs[31]; IL-4 has been reported to suppress enzymes required for CLA synthesis, especially FUT7[32,33]. Finally TGF-β would enhance FUT7 expression in T cells[34]. However, cytokine stimulation induced only a mild, transient CLA decrease on day 4 under high-dose IL-12 and IL-4, with no effect of TGF-β, partially contradicting prior reports (figures 4.D-E). We also tested whether nutrient depletion or metabolite accumulation at high density contributed to reduced CLA. Daily media exchange experiments showed only minimal effects, with no rescue of CLA levels (figure 4.F). Expression analysis of key enzymes involved in CLA biosynthesis (FUT7, B4GALT1, GCNT1, FUT4, ST3GAL4 and B3GNT3[15,35]) showed no consistent downregulation accompanying CLA loss (figure S13). In some cases (e.g., FUT7 in OCI-AML2), expression even slightly increased at higher density (figure 4.G). GCNT1 and B3GNT3 were not reliably detected. Finally, given that CLA is an adhesion-related glyco-epitope and adhesion

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molecules are often density-regulated, we assessed multiple adhesion markers but observed no correlation with CLA reduction (figure S14). In summary, CLA is consistently downregulated at high cell density across cell types, but the underlying mechanism remains unresolved.

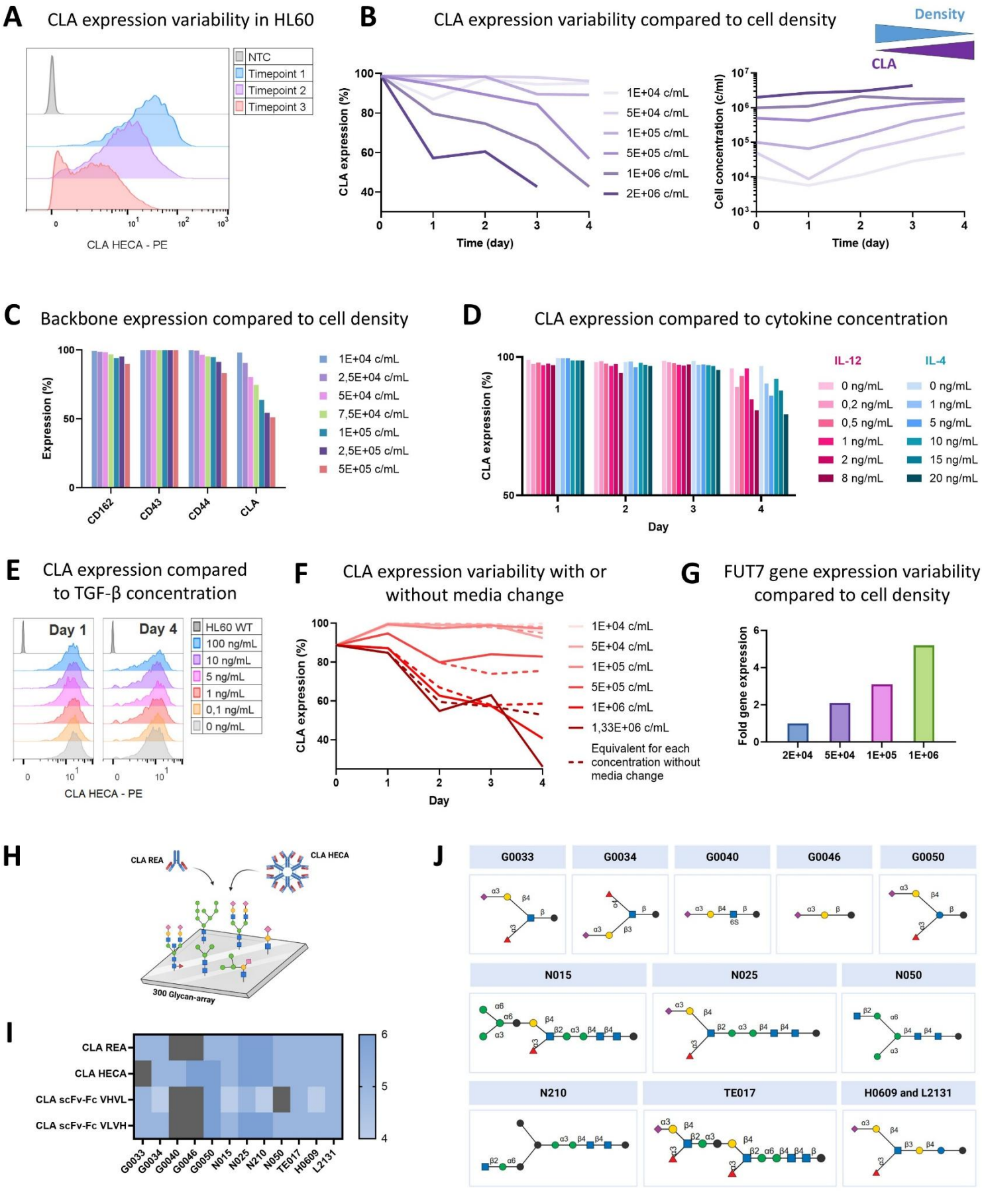
Broad and heterogeneous glycan recognition by CLA binders challenge the current definition of CLA

As previously noted, contradictory findings in the literature and our study suggest that CLA may be more complex than originally described, and that antibody specificity may be limited. We therefore profiled glycan binding using a 300-glycan array with REA, HECA, and scFv-Fc binders to define CLA-recognized structures (figure 4.H). The canonical CLA structure corresponds to glycan G0033 in this assay. Unexpectedly, all four binders interacted significantly with multiple glycans beyond G0033, and HECA did not significantly bind G0033 (Sialyl Lewis X). In total, 12 glycans were recurrently recognized, although binding profiles varied between binders (figures 4.I-J and S15). Among these, several matched Sialyl Lewis X-related motifs with extended structures (N025, TE017, H0609, L2131) or Sialyl Lewis A-like features (G0034). Others shared partial or truncated motifs (G0050, G0040, G0046), while some (N015, N210) displayed more divergent glycan architectures. Overall, these data suggest either substantial antibody cross-reactivity or that CLA should be defined as a broader glycan family rather than a single epitope.

Figure 4. Investigation of density-dependent CLA variability and its underlying regulatory mechanisms. ***A)*** *Flow cytometry analysis of CLA expression variability in HL60 cells at different time points during culture.* ***B)*** *Flow cytometry analysis of CLA expression (%) (left) and cell concentration (cells/mL) (right) over time in HL60 cells cultured at starting densities ranging from 1 × 10^4 to 2 × 10^6 cells/mL. Representative of a technical duplicate shown.* ***C)*** *Flow cytometry analysis of CD162, CD43 and CD44 expression across cell densities (1 × 10^4 to 5 × 10^5 cells/mL) compared with CLA expression in HL60.* ***D)*** *Flow cytometry analysis of CLA expression over time in HL60 cells cultured with varying IL-12 (0-8 ng/mL) and IL-4 (0-20 ng/mL) concentrations.* ***E)*** *Flow cytometry analysis of CLA expression on day 1 and day 4 of HL60 cells cultured with varying TGF-β concentrations (0-100 ng/mL).* ***F)*** *Flow cytometry analysis of CLA expression over time in HL60 cells cultured at different cell densities (1 × 10^4 to 1.33 × 10^6 cells/mL). From day 1 onward, half of the cultures were maintained unchanged, while the remaining cultures received daily medium replacement (dotted lines) to assess the effects of nutrient depletion and metabolite accumulation.* ***G)*** *Bar plot showing fold change in FUT7 expression measured by qRT-PCR in OCI-AML2 cells after 5 days of culture at different cell densities (2 × 10^4 to 1 × 10^6 cells/mL).* ***H)*** *Schematic of a 300-glycan array using CLA REA and CLA HECA clones to identify glycan-binding specificities and characterize CLA-associated glyco-structures. Created with BioRender.com.* ***I)*** *Heatmap of the 12 top glycan candidates showing significantly increased binding (z-score-based) for one or several CLA binders. Each sample was tested in triplicates.* ***J)*** *Structures of glycan components corresponding to the 12 top binding candidates. Structures were generated using the GlycoGlyph Toolkit[62] based on IUPAC-condensed glycan nomenclature. Full glycan names and plate layout are provided in the supplementary materials. Created with BioRender.com.*

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Figure 4



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4 Discussion

PDAC remains a devastating disease with limited therapeutic options. However, immunotherapy is opening new avenues, with several promising targets emerging, including CD318, already advancing toward clinical evaluation[36]. Beyond CD318, additional targets such as mesothelin[37], CEA[38], claudin 18.2[39] and HER2[40] have been explored, but their expression in normal tissues has raised safety concerns and, in some cases, caused severe adverse events, including fatal systemic toxicity in HER2 CAR T cells due to low-level pulmonary expression[41] and transient colitis in CEA-directed CAR T cells[42]. Within this landscape, we propose CLA as an attractive PDAC target due to its restricted expression in tumor tissue and selected leukocyte subsets[9]. Importantly, we confirmed the absence of detectable expression in healthy tissues, reinforcing its therapeutic interest. The only exception was the kidney cortex, where the observed staining remains unexplained and may represent either secreted antigen binding or nonspecific antibody reactivity, warranting further investigation. Beyond PDAC, CLA has been implicated in lymphoma[43] and skin inflammatory diseases[44], highlighting broader biological relevance.

Engineering a functional CLA CAR proved challenging due to the non-standard nature of the antigen. Although uncommon, glycan-targeting CARs have been described, as aberrant glycosylation is a hallmark of cancer[45]. CAR T cells targeting Tn-MUC1 have demonstrated efficacy in leukemia, pancreatic and breast cancer models[46] and are in phase I evaluation[47]. GD2-directed CARs have shown strong clinical potential across neuroblastoma[48], glioblastoma[49], sarcoma and melanoma[50]. Finally, preclinical studies have shown Lewis Y CARs efficacy in prostate cancer models[51]. In PDAC, altered glycosylation, including high-mannose glycans, is prominent, and lectin-based H84T BanLec CAR T cells have shown potent preclinical activity[52,53]. These studies collectively support glyco-epitopes as viable CAR targets, conceptually aligned with CLA.

However, engineering glycan-targeting CARs remains challenging due to difficulties in generating high-affinity and specific binders. Antibody development against glycans is often limited by weak immunogenicity and tolerance to human glyco-structures in murine systems[45]. For CLA, we relied on its natural HECA-452 IgM binder and derived IgG and scFv formats. To improve CAR performance, we tested common failure modes, including hinge optimization or tag removal, but this was insufficient[9]. We therefore investigated self-binding, misfolding and affinity limitations. Self-binding was prevented by FUT7 KO, resulting in CLA depletion on T cells; however, this did not restore the cytotoxic activity of the CLA CAR. Potential misfolding caused by the Myc and His tags was investigated, but tag removal likewise failed to improve killing. To assess the functionality of the scFv itself, scFv-Fc fusion proteins were generated and used in the AdCAR system. Their compatibility with the AdCAR platform confirmed that the scFv remained functional. Although a global folding issue of the complete CAR construct cannot be excluded, the data increasingly pointed toward an affinity-related limitation.

Affinity maturation using *in silico* and YSD approaches identified multiple candidates but failed to generate functional CARs. Nevertheless, we successfully adapted a glycan-compatible YSD workflow using HL60 cells with WT/KO dual selection, broadening the applicability of the method. Alternative approaches such as phage display could offer broader sequence exploration, while yeast display remains superior for fine affinity tuning. However, neither strategy addresses higher-order

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CAR constraints such as folding, orientation or dimerization. CAR pooling or functional screening approaches (based on CAR activity rather than binder affinity) may better capture these effects in future studies[54]. *In silico* affinity maturation generated multiple mutations but no functional CARs, likely due to model limitations trained on full antibodies rather than scFvs and lack of structural constraints[24]. Many candidates were germline-like substitutions with uncertain benefit. Emerging models such as BoltzGen[55], OrthoRep[56] or RFdiffusion[57] may overcome these limitations.

Despite excluding individual failure modes, no functional CAR initially emerged, suggesting combined interdependent constraints rather than a single dominant cause. We therefore transitioned to an AdCAR strategy, which overcomes several limitations of direct CARs such as dependence on antigen density and epitope geometry. AdCARs exploit multivalent adapter-mediated interactions, enhancing avidity and flexibility, while scFv-Fc adapters improve valency and stability, potentially reducing misfolding. Using biotinylated CLA scFv-Fc adapters, we achieved robust dose-dependent cytotoxicity, validating CLA as a functional PDAC target for the first time. Importantly, efficient killing occurred without prior FUT7 KO, suggesting limited self-antigen impact. Another noteworthy observation was that the scFv-Fc format required approximately 10-fold higher adapter concentrations than the REA format to achieve comparable cytotoxicity, despite that the REA had lower titrated binding performance. These stochiometric differences could be explainable by the multitude of parameters playing into the functionality of the AdCAR system. While affinities may be different between the different antibody variants, the degree of biotinylation might as well. As we did not investigate the influence of the degree of biotinylation, it is hard to pin down the exact reason for discrepancies in the binding affinity and killing. While this finding warrants further characterization in the context of preclinical development, it did not impair CAR function here.

Beyond engineering aspects, this work also provides new insights into CLA biology. We confirm FUT7 as essential for CLA expression and its knockout does not impair T cell function. In contrast, CD162, CD44 or CD43 KO did not abolish CLA expression, indicating that additional, as yet unidentified, CLA carriers exist. These carriers may include other glycoproteins or, potentially, non-protein molecules such as glycosphingolipids, which have been reported to carry carbohydrate structures related to CLA, including the sialyl Lewis X motif[58]. Proteomic identification was inconclusive, likely due to nonspecific binding or potentially dynamic redistribution of CLA across multiple scaffolds. Glycan arrays revealed large and heterogeneous binding of CLA binders, supporting either differential recognition or a broader CLA glyco-epitope family. The implication of this finding is that our previous analysis of the HECA clones may not be entirely accurate, as we show that this binding structure exhibits promiscuous binding properties.

We also identified strong cell density-dependent regulation of CLA across multiple cell types. We performed extensive investigation, including cytokine modulation (IL-12, IL-4, TGF-β), protein expression (backbone and adhesion molecule regulation) or metabolic modulation (nutrient depletion, metabolic product accumulation and enzymes regulation). However, we could not identify a single responsible mechanism. This could suggest a multifactorial or context-dependent regulation that would need further investigation. Future studies could also be extended by performing cell cycle analyses or by examining the expression and activity of enzymes involved in CLA degradation or, more broadly, glycosphingolipid metabolism, as the expression of these enzymes is often influenced by cell density[59–61].

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In conclusion, this study demonstrates that CLA is a viable PDAC CAR target when implemented via AdCAR, overcoming intrinsic limitations of direct CAR design. We further uncover density-dependent regulation and incomplete structural definition of CLA. While further work is needed to fully resolve its molecular identity and regulatory network, CLA represents a promising target not only for cancer immunotherapy but potentially also for inflammatory and dermatological diseases.

5 Data Availability Statement

The datasets used and/or analyzed in this paper are available from the corresponding authors on reasonable request. The glycol-array raw data and analyzed results are available in supplementary files.

6 Ethics statement

PBMCs were isolated from the whole blood of healthy anonymous volunteers at Miltenyi Biotec, with written informed consent as approved by the local ethics committee of Ärztekammer Nordrhein (2020272). All blood samples were handled following the required ethical and safety procedures.

7 Author Contributions

C.D and D.S conceived the study and developed the experimental strategies. C.D performed the staining, killing assays, the triple CD43/CD44/CD162 KO, the CLA affinity maturation, the AdCAR assay, the CLA regulation investigations and the CLA glycan analysis. C.D further performed the data analysis and the manuscript preparation. L.G and D.S performed the CLA direct-CAR designs, the FUT7 and CD162 KO investigations and analysis. D.S performed the cyclic immunofluorescence staining and analysis. K.V supported C.D for the cytotoxicity assay handling and manuscript revision. N.B, M.D and A.F supported C.D for the affinity maturation by yeast display. S.T supported the glyco-array analysis and investigated CLA backbones and glycosylation enzymes. H.H and J.W produced the anti-CLA scFv-Fc molecules. D.S and O.H contributed to supervision. All authors contributed to manuscript revision.

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lentivectors; Aleksej Frolov and Camille Métais for glycan array analysis; Anniek Timmermans and Dave Huang for their contributions to the project as master's students.

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10 Conflict of Interest

C.D, K.V, N.B, M.D, A.F, J.W, S.T, H.H, O.H and D.S are employees of Miltenyi Biotec B.V. & Co. KG. L.G was an employee of Miltenyi Biotec B.V. & Co. KG at the time the experiments were performed. All other authors declare no competing interests.

11 Generative AI statement

The author(s) declare that ChatGPT (GPT-5.3, OpenAI) was used in the revision of this manuscript, to improve the readability and language. After using this tool, the authors reviewed the content critically and thoroughly, edited it wherever needed, and take full responsibility for the content of the publication. All scientific content, interpretations and conclusions were generated and verified by the authors.

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Supplementary tables:

Supplementary table 1. List of Miltenyi Biotec antibodies used for flow cytometric analysis, cyclic immunofluorescence and CLA carrier protein identification.

Target	Clone	Conjugate
AdCAR detection reagent	na	PE
Anti-biotin	REA746, human	PE
Anti-HA	REA1288, human	APC
Anti-His	GG11-8F3.5.1, mouse	FITC and PE
Anti-Myc	SH1-26E7.1.3, mouse	FITC
CLA HECA-452	HECA-452, rat IgM	PE and unconjugated and biotinylated
CLA REA	REA1101, human IgG1	PE and unconjugated
CD2	REA972	PE-Vio770
CD4	REA623, human	VioGreen
CD8	REA734, human	VioBlue
CD11a	REA378, human	PE
CD25	REA570, human	PE-Vio770
CD43	REA833, human	VioGreen
CD44	REA690, human	VioBlue
CD45	REA747, human	Unconjugated and biotinylated
CD45RA	REA562, human	VioBlue
CD45RO	REA611, human	APC
CD49d	REA545, human	PE
CD54 (ICAM-1)	REA266, human	APC
CD58 (LFA-3)	REA1098, human	PE-Vio770
CD62L	REA828, human	PE-Vio770
CD66c	REA414, human	Unconjugated and biotinylated
CD69	REA824, human	APC-Vio770
CD95	REA738, human	PE
CD106 (VCAM-1)	REA269, human	PE-Vio770
CD137 (4-1BB)	REA765, human	APC

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CD162 (PSGL1)	REA319, human	APC, unconjugated and biotinylated
CD183 (CXCR3)	REA232, human	PE
CD184 (CXCR4)	REA649, human	APC-Vio770
CD197 (CCR7)	REA108, human	APC
CD223 (LAG3)	REA351, human	VioBlue
CD271 (LNGFR)	REA844, human	APC
CD279 (PD1)	REA1165, human	PE-Vio770
CD326 (Epcam)	REA764, human	FITC

Supplementary table 2. List of mutations identified through in silico and in vitro affinity maturation. *For each mutation, the original amino acid class and the substituted amino acid class are indicated. The source of the prediction is specified as either in silico (AI), including the number of models supporting the mutation (1 to 6), or in vitro using yeast surface display (YSD). The location of each mutation is annotated as belonging to either a framework region or a complementarity-determining region (CDR). Information on whether each mutation corresponds to a germline residue (yes/no) is provided, according to IMGT/V-QUEST. Mutations at germline positions but involving a different amino acid labelled as "different substitution".*

Mutation	Original Class	Substituted Class	Source of prediction	Region	Germline mutation
N16G	Polar	Non-polar	AI - 6	Framework	Yes
E43K	Acidic	Basic	AI - 6	Framework	Yes
D46E	Acidic	Acidic	AI - 6	Framework	Yes
A65G	Non-polar	Non-polar	AI - 6	Framework	Yes
L82M	Non-polar	Non-polar	AI - 6	Framework	Yes
T166P	Polar	Non-polar	AI - 6	Framework	Yes
V111T	Non-polar	Polar	AI - 5	Framework	Yes
M120T	Non-polar	Polar	AI - 4	Framework	Different substitution
Q124S	Polar	Polar	AI - 4	Framework	Different substitution
V58Y	Non-polar	Polar	AI - 3	Framework	Yes
I185S	Non-polar	Polar	AI - 3	Framework	Yes
W129V	Non-polar	Non-polar	AI - 3	Framework	Different substitution
A168L	Non-polar	Non-polar	AI - 2	CDR5	Yes
Q118A	Polar	Non-polar	AI - 2	Framework	Different substitution
E198S	Acidic	Polar	AI - 2	Framework	Yes
W129S	Non-polar	Polar	AI - 2	Framework	Yes
V175T	Non-polar	Polar	AI - 2	CDR5	Different substitution
A125P	Non-polar	Non-polar	AI - 2	Framework	Yes
A98D	Non-polar	Acidic	AI - 1	CDR3	Yes
M112T	Non-polar	Polar	AI - 1	Framework	Yes
M120V	Non-polar	Non-polar	AI - 1	Framework	Yes
W129T	Non-polar	Polar	AI - 1	Framework	Different substitution
H153T	Basic	Polar	AI - 1	CDR4	Yes
V175S	Non-polar	Polar	AI - 1	CDR5	Different substitution

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S196T	Polar	Polar	AI - 1	Framework	Different substitution
L211A	Non-polar	Non-polar	AI - 1	CDR6	Different substitution
H172W	Basic	Non-polar	AI - 1	CDR5	Different substitution
D178S	Acidic	Polar	AI - 1	Framework	Yes
T182D	Polar	Acidic	AI - 1	Framework	Yes
M112L	Non-polar	Non-polar	AI - 1	Framework	Yes
S30A	Polar	Non-polar	YSD	CDR1	No
S52V	Polar	Non-polar	YSD	CDR2	No
A96V	Non-polar	Non-polar	YSD	CDR3	No
Q144R	Polar	Basic	YSD	CDR4	No
A168E	Non-polar	Acidic	YSD	CDR5	Different substitution
H172P	Basic	Non-polar	YSD	CDR5	Different substitution
L147G	Non-polar	Non-polar	YSD	CDR4	No
I170H	Non-polar	Basic	YSD	CDR5	No
S216G	Polar	Non-polar	YSD	CDR6	Not same AA

Supplementary table 3. List of affinity-matured CLA CAR variants containing mutations identified through in silico and in vitro affinity maturation. *Due to the large number of predicted mutations and the limited throughput of CAR cloning and functional assays, 40 candidate combinations were selected for testing. All in silico mutations supported by at least four prediction models were included. In addition, selected mutations supported by three, two or one model(s) were incorporated, preferentially within CDR regions, to ensure representation across confidence categories. A CAR containing all the 12 mutations supported by at least three models was also generated. All mutations identified by YSD were tested individually, in combination and in "minus-one" combinations to identify mutations critical for affinity improvement. Because clone C2 contained a truncated heavy chain, all corresponding C2-derived combinations were additionally evaluated in a truncated heavy-chain CLA CAR format.*

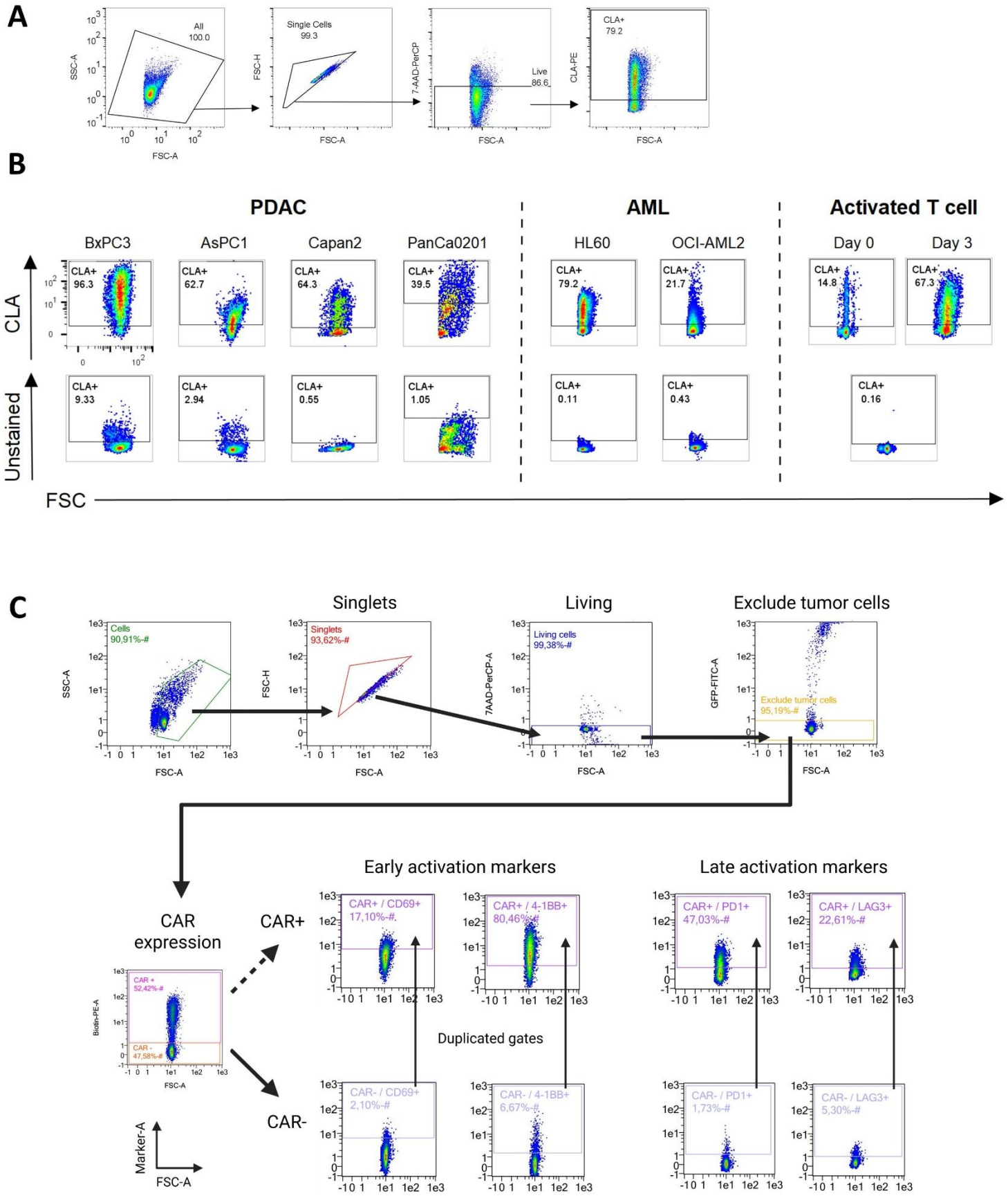
Affinity matured CAR	Source of prediction	Mutation(s)	Description
1	AI (*in silico*)	N16G	Supported by 6 models
2	AI (*in silico*)	E43K	Supported by 6 models
3	AI (*in silico*)	D46E	Supported by 6 models
4	AI (*in silico*)	A65G	Supported by 6 models
5	AI (*in silico*)	L82M	Supported by 6 models
6	AI (*in silico*)	T166P	Supported by 6 models
7	AI (*in silico*)	V111T	Supported by 5 models
8	AI (*in silico*)	M120T	Supported by 4 models
9	AI (*in silico*)	Q124S	Supported by 4 models
10	AI (*in silico*)	V58Y	Supported by 3 models
11	AI (*in silico*)	W129V	Supported by 3 models
12	AI (*in silico*)	A168L	Supported by 2 models
13	AI (*in silico*)	V175T	Supported by 2 models
14	AI (*in silico*)	A98D	Supported by 1 model
15	AI (*in silico*)	V175S	Supported by 1 model
16	AI (*in silico*)	L211A	Supported by 1 model
17	AI (*in silico*)	H172W	Supported by 1 model
18	AI (*in silico*)	N16G, E43K, D46E, A65G, L82M, T166P, V111T, M120T, Q124S, V58Y, I185S, W129V	All 12 mutations supported by at least 3 models combined

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19	YSD (*in vitro*)	S30A, S52V, A96V, Q144R, A168E, H172P	All 6 mutations supported by clone D10
20	YSD (*in vitro*)	S52V, A96V, Q144R, A168E, H172P	5 mutations supported by clone D10 without S30A
21	YSD (*in vitro*)	S30A, A96V, Q144R, A168E, H172P	5 mutations supported by clone D10 without S52V
22	YSD (*in vitro*)	S30A, S52V, Q144R, A168E, H172P	5 mutations supported by clone D10 without A96V
23	YSD (*in vitro*)	S30A, S52V, A96V, A168E, H172P	5 mutations supported by clone D10 without Q144R
24	YSD (*in vitro*)	S30A, S52V, A96V, Q144R, H172P	5 mutations supported by clone D10 without A168E
25	YSD (*in vitro*)	S30A, S52V, A96V, Q144R, A168E	5 mutations supported by clone D10 without H172P
26	YSD (*in vitro*)	L147G, I170H, S216G	All 3 mutations supported by clone C02
27	YSD (*in vitro*)	I170H, S216G	2 mutations supported by clone C02 except L147G
28	YSD (*in vitro*)	L147G, S216G	3 mutations supported by clone C02 except I170H
29	YSD (*in vitro*)	L147G, I170H	4 mutations supported by clone C02 except S216G
30	YSD (*in vitro*)	L147G, I170H, S216G	All 3 mutations supported by clone C02 with the light chain only
31	YSD (*in vitro*)	S30A	Single mutation supported by clone D10
32	YSD (*in vitro*)	S52V	Single mutation supported by clone D10
33	YSD (*in vitro*)	A96V	Single mutation supported by clone D10
34	YSD (*in vitro*)	Q144R	Single mutation supported by clone D10
35	YSD (*in vitro*)	A168E	Single mutation supported by clone D10
36	YSD (*in vitro*)	H172P	Single mutation supported by clone D10
37	YSD (*in vitro*)	L147G	Single mutation supported by clone C2
38	YSD (*in vitro*)	I170H	Single mutation supported by clone C2
39	YSD (*in vitro*)	S216G	Single mutation supported by clone C2
40	na	na	CLA parental clone (WT)

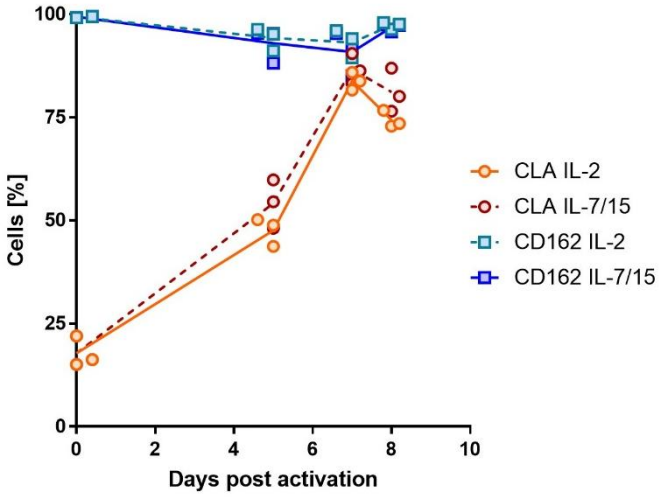
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Supplementary figures:



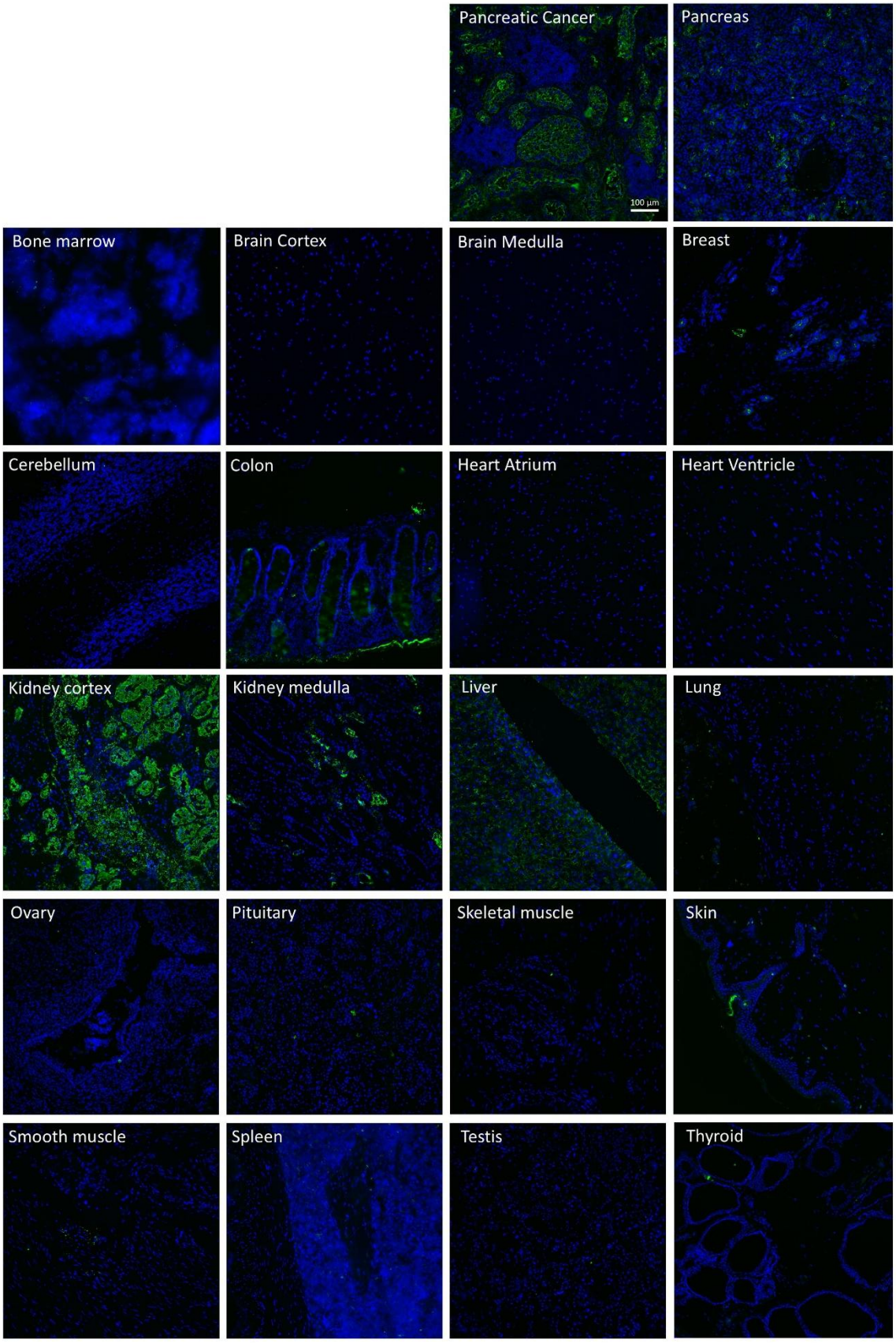
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Supplementary figure S1. Flow cytometry gating strategies.
A) *Gating strategy for single-marker analysis (e.g., CLA). Initial gating steps excluded debris (FSC-A vs SSC-A), doublets (FSC-A vs FSC-H) and dead cells (FSC-A vs 7AAD/PerCP-A). The marker of interest was gated relative to unstained control (NTC).* ***B)*** *Gating strategy for CLA analysis in PDAC and leukemia cell lines, as well as T cells, compared with their respective unstained controls (NTC).* ***C)*** *Gating strategy for activation marker analysis following CLA AdCAR killing assays. Initial gating steps excluded debris (FSC-A vs SSC-A), doublets (FSC-A vs FSC-H) and dead cells (FSC-A vs 7AAD/PerCP-A). CAR^+ T cells were identified as biotin-PE^+. Early activation was assessed by CD69 and 4-1BB expression, and late activation by PD-1 and LAG-3 expression. Gates were defined using CAR^- cells and the non-transduced control (NTC), then applied to CAR^+ cells to ensure unbiased thresholding across T-cell populations. Created with BioRender.com.*



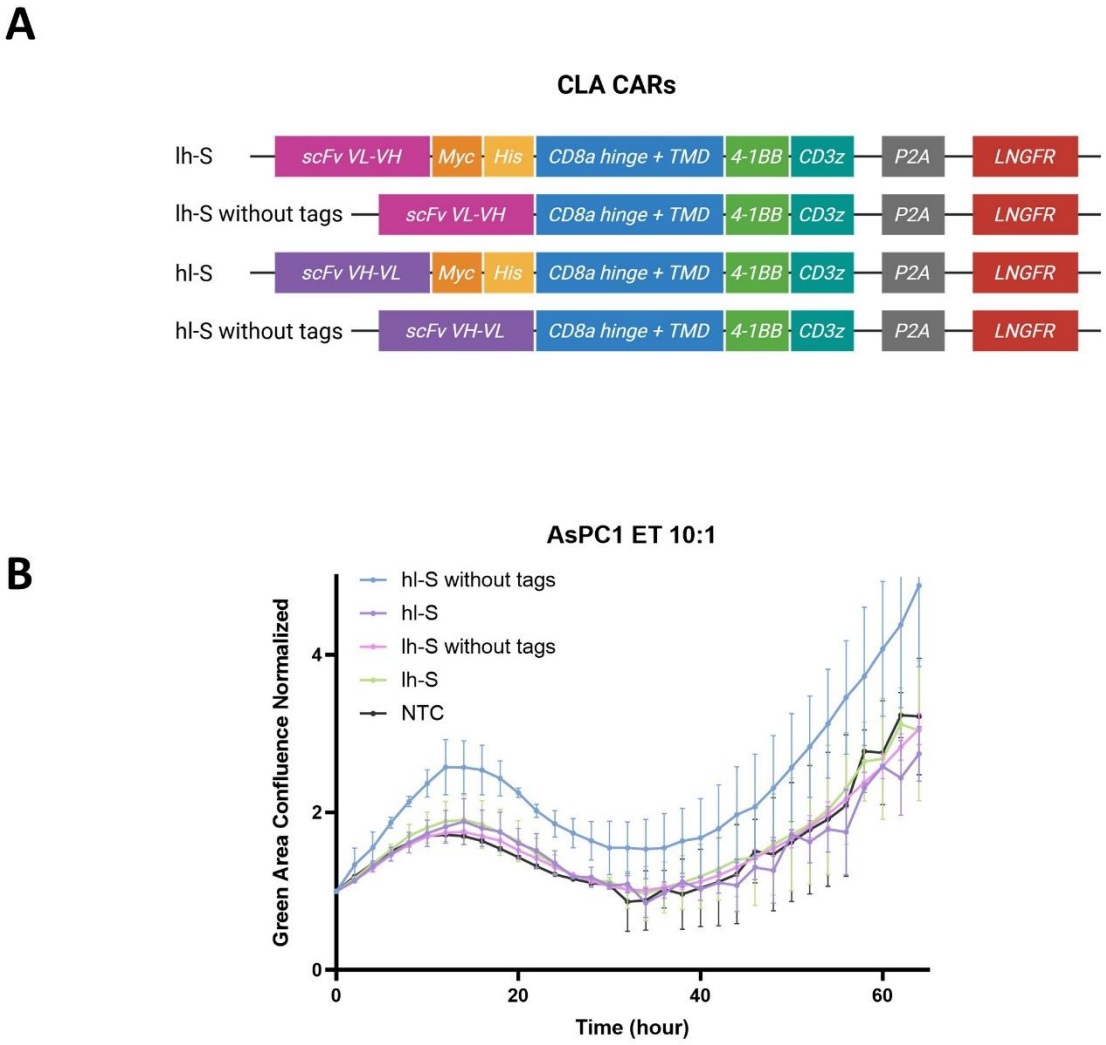
Supplementary figure S2. Flow cytometry analysis of CLA and CD162 expression on T cells following activation with TransAct™ over time. *Data represents the mean of T cells from three independent donors cultured in medium supplemented with either IL-2 (12.5 ng/mL) or IL-7 and IL-15 (12.5 ng/mL each).*

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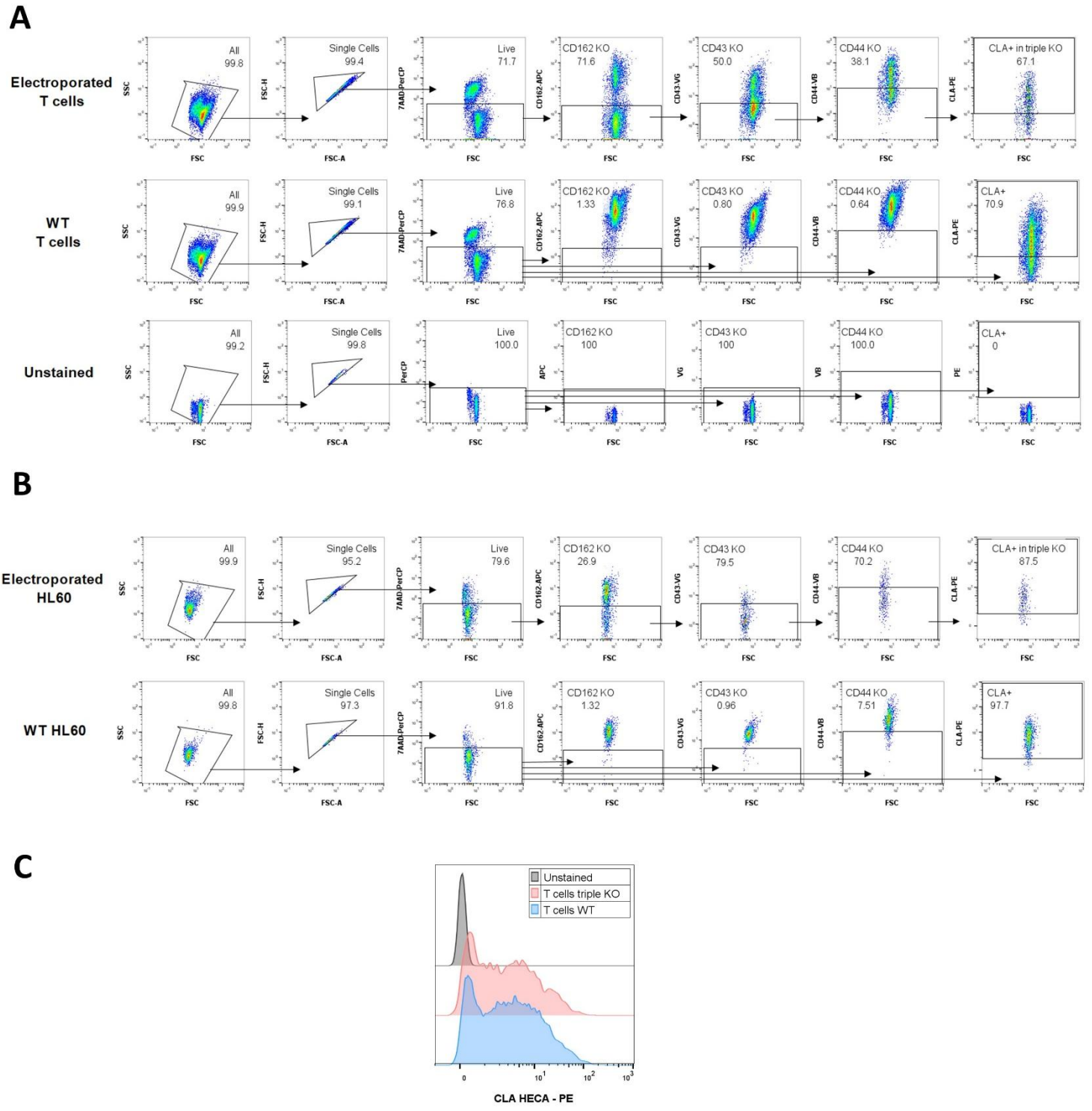
Supplementary figure S3. CLA expression on healthy tissues assessed by cyclic immunofluorescence imaging. *Representative cyclic immune fluorescence images of several healthy tissues stained with an anti-CLA-PE conjugate (HECA-452). Scale bar = 100 µm. Images are representative of at least two regions of interest from one tissue. Regions of interest were chosen based on manual DAPI and Cytokeratin prestaining and dependency on the respective tissue size.*

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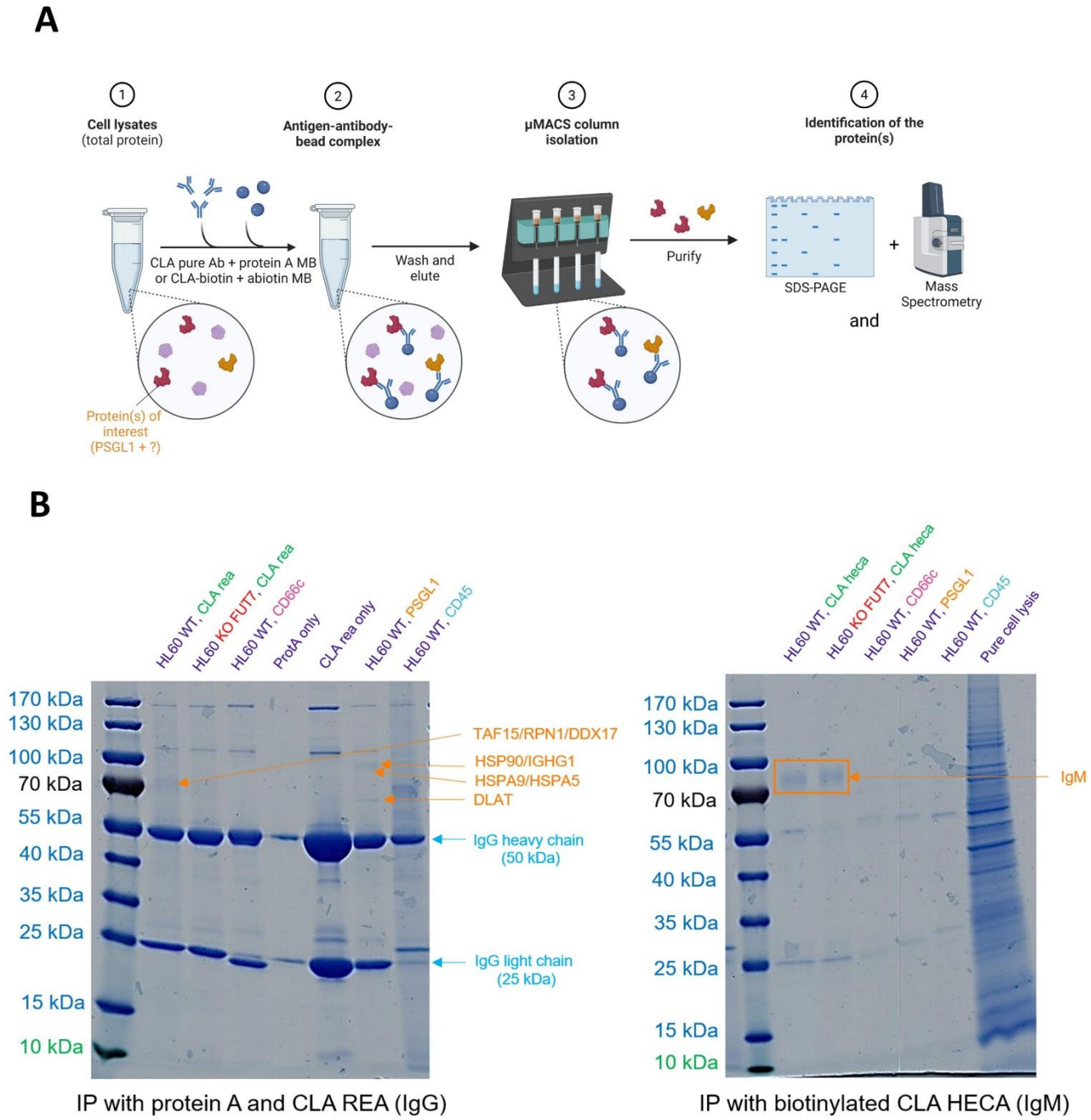
Supplementary figure S4. Test of the functionality of the initial CLA CAR designs. A) *Schematic of the second-generation CLA CAR cassette containing two scFv orientations (VLVH or VHVL, referred to as lh and hl respectively), a short CD8α hinge (S), a transmembrane domain (TMD), a 4-1BB co-stimulatory domain, a CD3ζ signaling domain, and optional Myc and His tags. The cassettes are followed by a Porcine teschovirus-1 2A self-cleaving peptide (P2A) and a low-affinity nerve growth factor receptor (LNGFR) as transduction marker. Created with BioRender.com.* ***B)*** *Killing curves showing GFP-based AsPC1 tumor cell confluence, normalized to the first time point, for hl-S and lh-S CLA CAR constructs with or without Myc and His tags at an E:T ratio of 10:1, compared with NTC. Error bars represent the mean of two independent donors.*

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Supplementary figure S5. CLA depletion through CRISPR/Cas9-mediated CD44, CD43 and CD162 triple KO in multiple cell lines. A-B) *Gating strategy for triple KO analysis in T cells* ***A****) and HL60 cells* ***B****). Initial gating excluded debris (FSC-A vs SSC-A), doublets (FSC-A vs FSC-H) and dead cells (FSC-A vs 7AAD/PerCP-A). CD162, CD43 and CD44 expression were gated relative to unstained (NTC) and the WT cells controls. Triple KO cells were identified sequentially by selecting CD162⁻ cells, followed by CD43⁻ cells, and finally CD44⁻ cells. CLA expression was then assessed within the CD162⁻CD43⁻CD44⁻ population. For control samples, each marker was analyzed relative to the total live-cell population rather than sequential parent gates.* ***C)*** *Flow cytometry analysis of CLA depletion in T cells following triple CD162/CD43/CD44 KO compared with WT.*

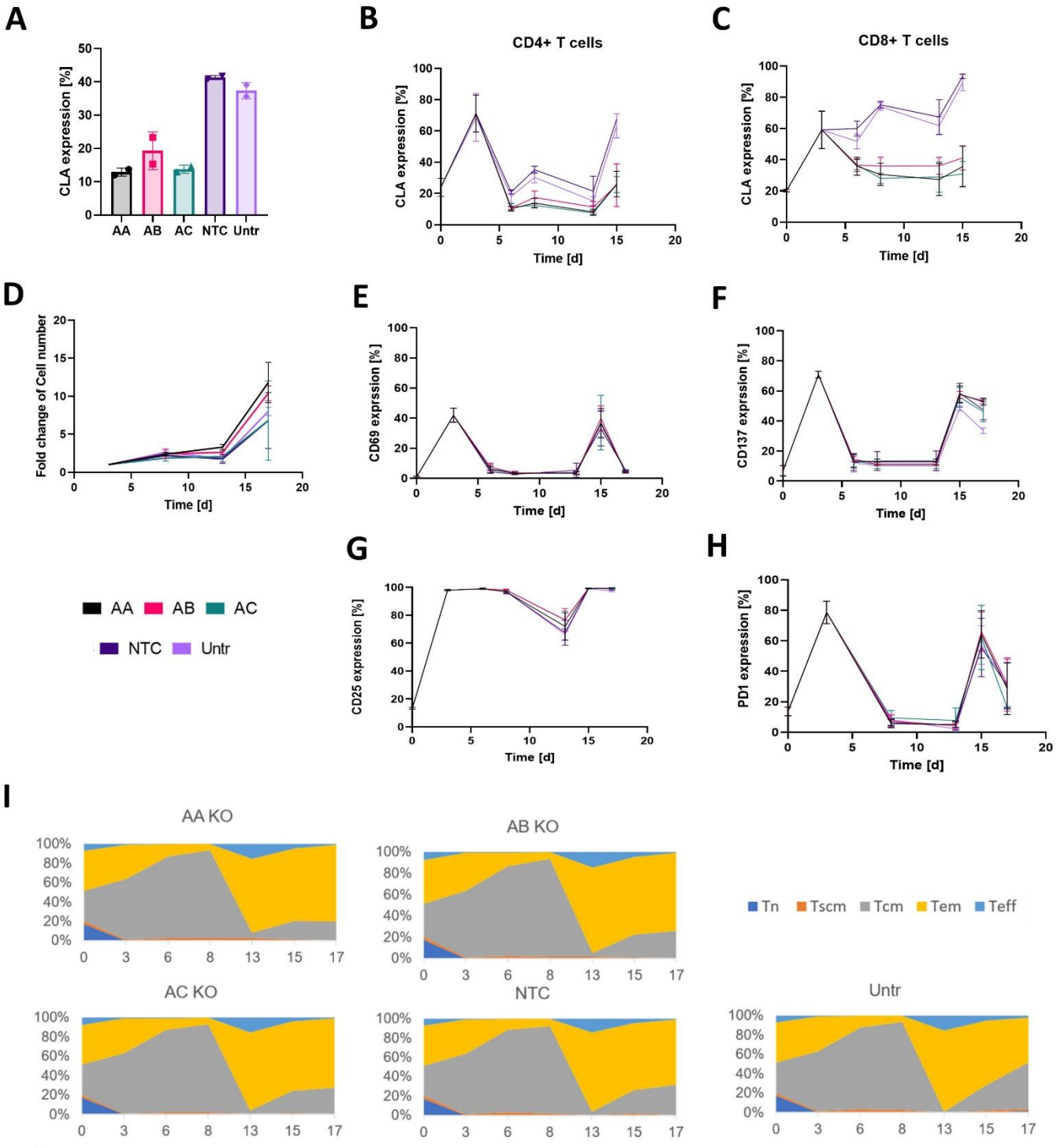
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Supplementary figure S6. Identification of CLA antibody carrier proteins in HL60 cells. ***A)*** *Workflow of the immunoprecipitation (IP) protocol. Two IP approaches were performed using HL60 cell lysates: (i) purified CLA REA antibody (IgG) coupled to Protein A, or (ii) biotinylated CLA HECA-452 antibody (IgM) coupled to anti-biotin microbeads. Immunoprecipitated proteins were analyzed by SDS-PAGE and identified by peptide mass fingerprint (PMF) analysis. Created with BioRender.com.* ***B)*** *SDS-PAGE analysis of IP samples from HL60 WT or CLA-depleted FUT7 KO cells. Left: IP performed with purified antibodies and Protein A. Right: IP performed with biotinylated antibodies and anti-biotin microbeads. Controls included CD66c antibody, Protein A only, CLA REA only, PSGL1/CD162 antibody, and CD45 antibody. Besides IgG-associated bands, the objective was to identify in HL60 WT cells treated with CLA antibodies: (i) a CD162/PSGL1 band comparable to the PSGL1/CD162 control, and (ii) additional bands potentially corresponding to alternative CLA carrier proteins that were absent in FUT7 KO cells. However, no clear candidate bands were detected. The only visible but weak band following CLA*

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REA IP was identified by PMF as TAF15, RPN1 and DDX17 (TATA-box binding protein-associated factor 15, ribophorin-1 and DEAD-box helicase 17), likely representing nonspecific binding artifacts rather than CLA carrier proteins. A ~60 kDa band observed in the PSGL1/CD162 condition was identified as DLAT (dihydrolipoamide S-acetyltransferase) by PMF analysis. Additional weak bands corresponded to heat shock proteins (HSPs) and IGHG1 (immunoglobulin heavy constant gamma 1). Following IP with biotinylated antibodies, no specific bands were detected apart from bands corresponding to the IgM antibody identified by PMF analysis.

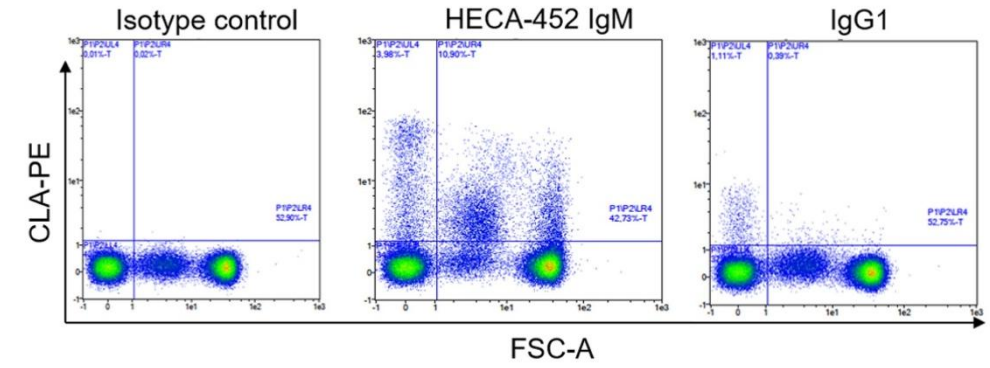


Supplementary figure S7. Analysis of FUT7 KO influence on T cell proliferation, markers and phenotype. ***A)*** *Bar plot showing CLA expression on peripheral T cells on day 8 after FUT7 KO using three sgRNAs (AA, AB, AC) compared with non-transduced electroporated control (NTC) and untreated control (Untr).*

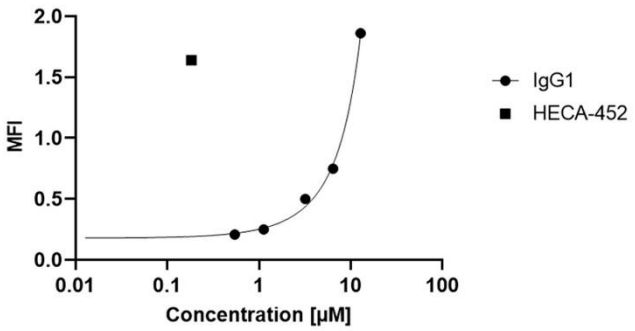
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B-C) *CLA expression over time on CD4⁺* ***B)*** *and CD8⁺* ***C)*** *T cells after FUT7 KO.* ***D)*** *Fold change in peripheral T cell counts over time following FUT7 KO.* ***E-H)*** *Expression of activation and exhaustion markers on T cells over time: CD69* ***E)****, CD137* ***F)****, CD25* ***G)****, and PD-1* ***H)*** *post-FUT7 KO.* ***I)*** *Area plots showing T cell differentiation after FUT7 KO. Percentages of T cell phenotypes over time were determined by flow cytometry based on CD62L, CD45RA and CD95 expression to assess differentiation status.* ***A-I)*** *Cells were activated with TransAct™ on days 0 and 13. All assays were performed in duplicate using blood from two independent donors. Tn: naïve T cells. Tscm: memory stem cell T cells. Tcm: central memory T cells. Tem: effector memory T cells. Teff: effector T cells.*

A



B



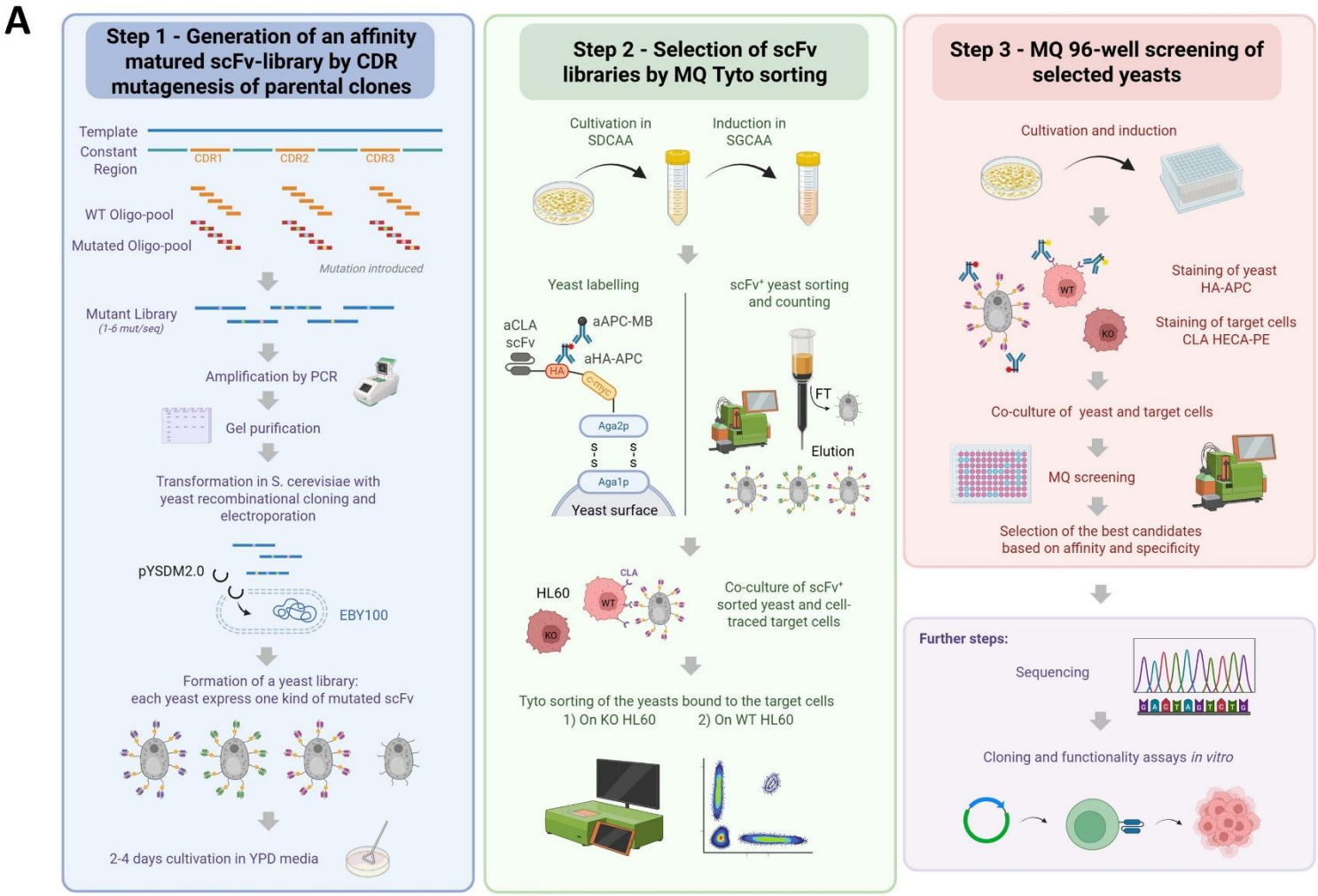
C

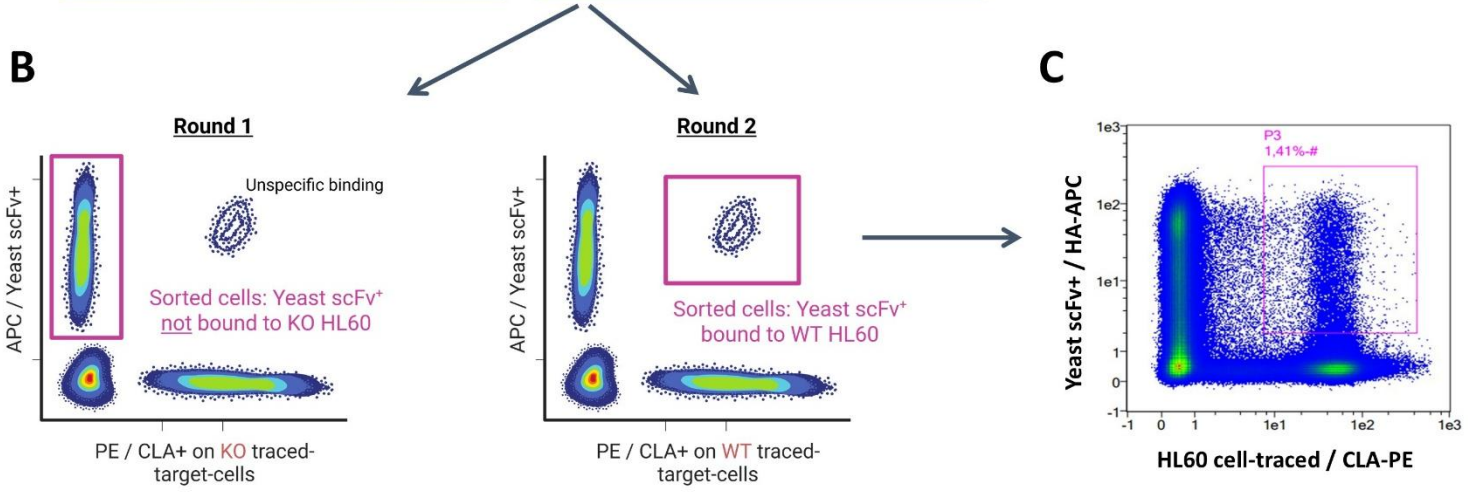
	Weight (Da)
HECA-452 IgM	900000
IgG	146402

	Given concentration [g/L]	Calculated molarity [uM]
HECA-452 IgM	0,165	0,18
IgG1 concentration 1	1,88	12,84
IgG1 concentration 2	0,94	6,42
IgG1concentration 3	0,47	3,21
IgG1 concentration 4	0,165	1,13
IgG-concentration 5	0,08	0,55

Supplementary figure S8. Titration of CLA REA IgG and HECA-452 IgM antibodies to compare staining performance. ***A)*** *Density plot of flow cytometry showing binding of 0.165 g/L CLA REA IgG or HECA-452 IgM antibodies on PBMCs.* ***B)*** *Graph of mean fluorescence intensity (MFI) across a range of antibody concentrations.* ***C)*** *Calculation of the molarity of the anti-CLA HECA-452 IgM and the IgG. The weight of the HECA-452 IgM is estimated based on similar antibodies. Both antibodies were evaluated for their staining performance toward CLA at the indicated concentrations. The respective molarity was calculated with the estimated weight.*

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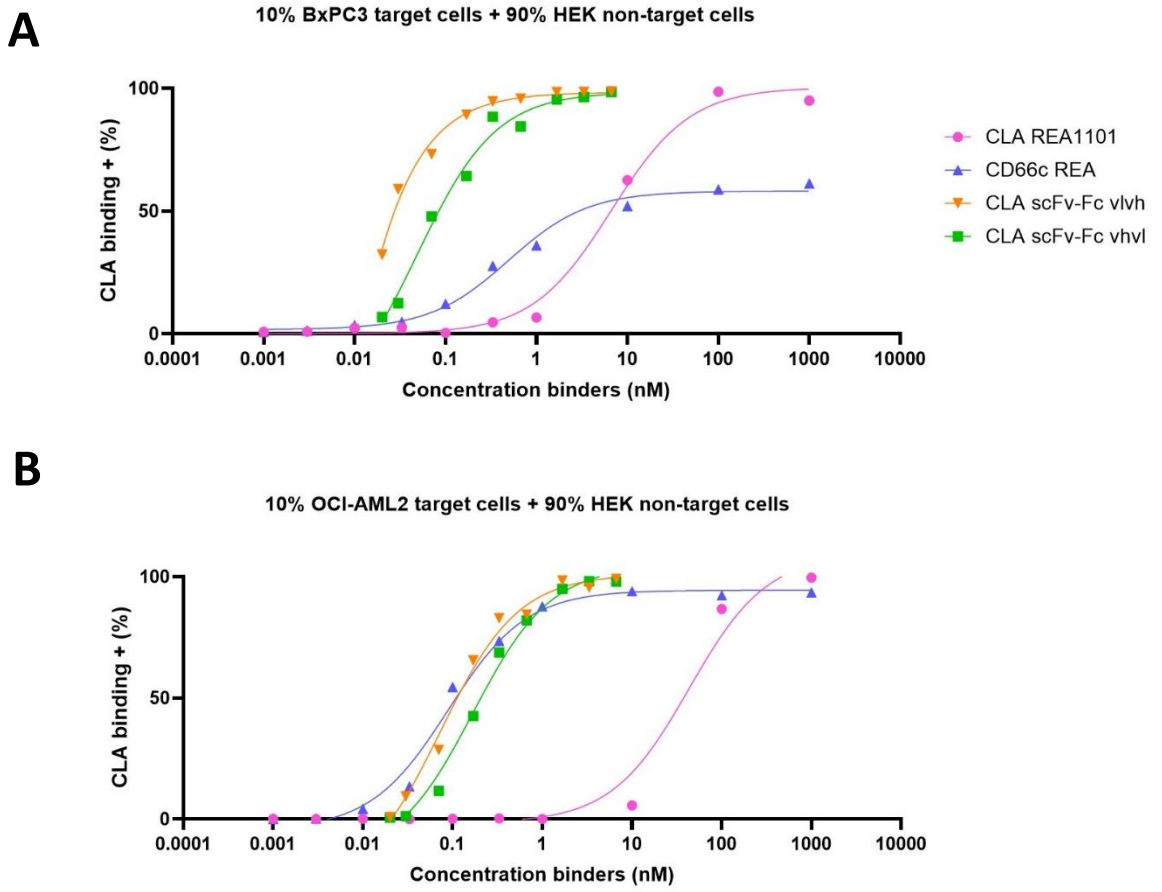






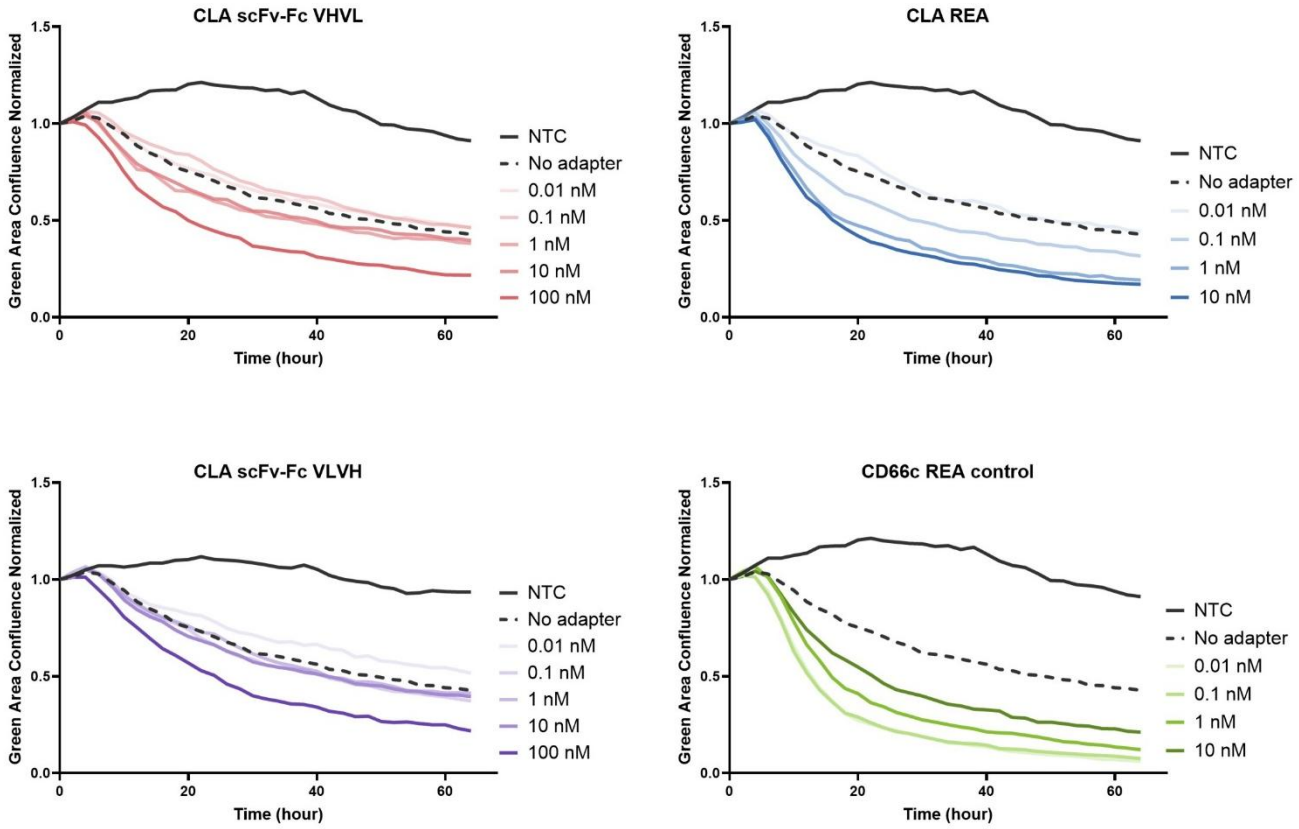
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Supplementary figure S9. Affinity maturation by yeast surface display (YSD). A) *Workflow of the in vitro YSD experiment.* ***B)*** *Schematic of the selection strategy during Tyto sorting. Round 1: HL60 KO cells are used to remove yeast expressing the HA tag that binds non-specifically. Round 2: HL60 WT cells are used to select yeast expressing the HA tag that specifically bind target cells.* ***A-B)*** *Created with BioRender.com.* ***C)*** *Flow cytometry analysis of yeast after Round 2 sorting. 1.41% of cells were selected, consistent with typical yields for this procedure.* ***D)*** *Sequencing results of two positive clones, D10 and C2, containing 6 and 3 mutations indicated with blue arrows, respectively. Clone C2 had a truncated heavy chain, so predicted mutations were only identified in the light chain.*



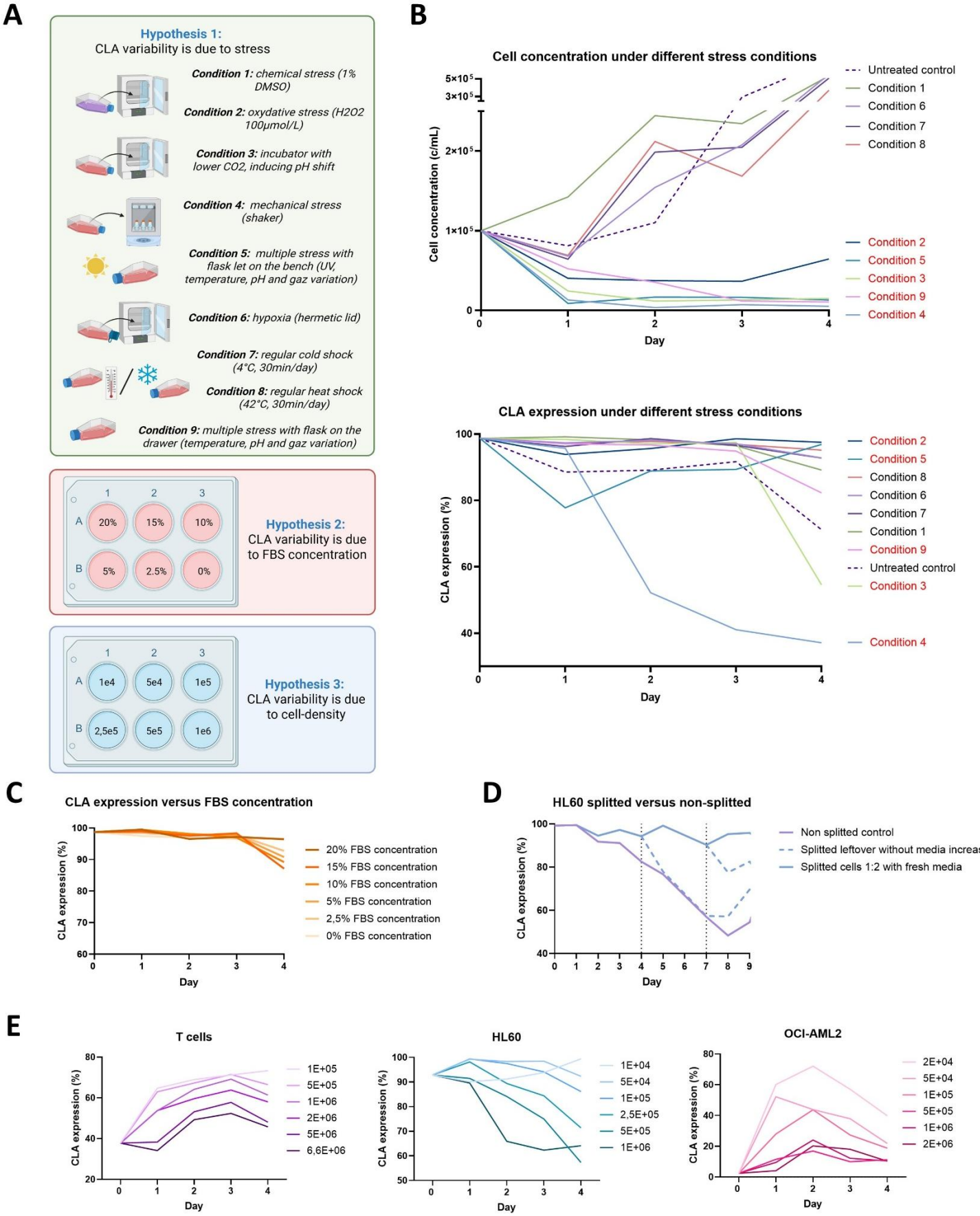
Supplementary figure S10. Binding performance of CLA scFv-Fc (VHVL or VLVH) compared with CLA REA IgG and CD66c REA IgG controls, evaluated by flow cytometry. *10% of target cells (A: BxPC3, B: OCI-AML2) were mixed with 90% non-target HEK cells. Data was analyzed using a three-parameter inhibitor-versus-response model with simple logistic regression in GraphPad Prism.*

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Supplementary figure S11. Killing curves of CLA scFv-Fc AdCAR assay for a second independent donor. *GFP-based BxPC3 tumor cell confluence, normalized to the first time point, is shown for CLA AdCAR constructs at an E:T ratio of 2:1 using: CLA scFv-Fc VHVL adapter, CLA scFv-Fc VLVH adapter, CLA REA adapter or CD66c REA adapter. Results are compared with non-transduced T cells (NTC) and AdCAR without adapter (no-adapter) controls. Biotinylated adapters were tested at 0.01-100 nM for scFv-Fc adapters and 0.01-10 nM for REA controls.*

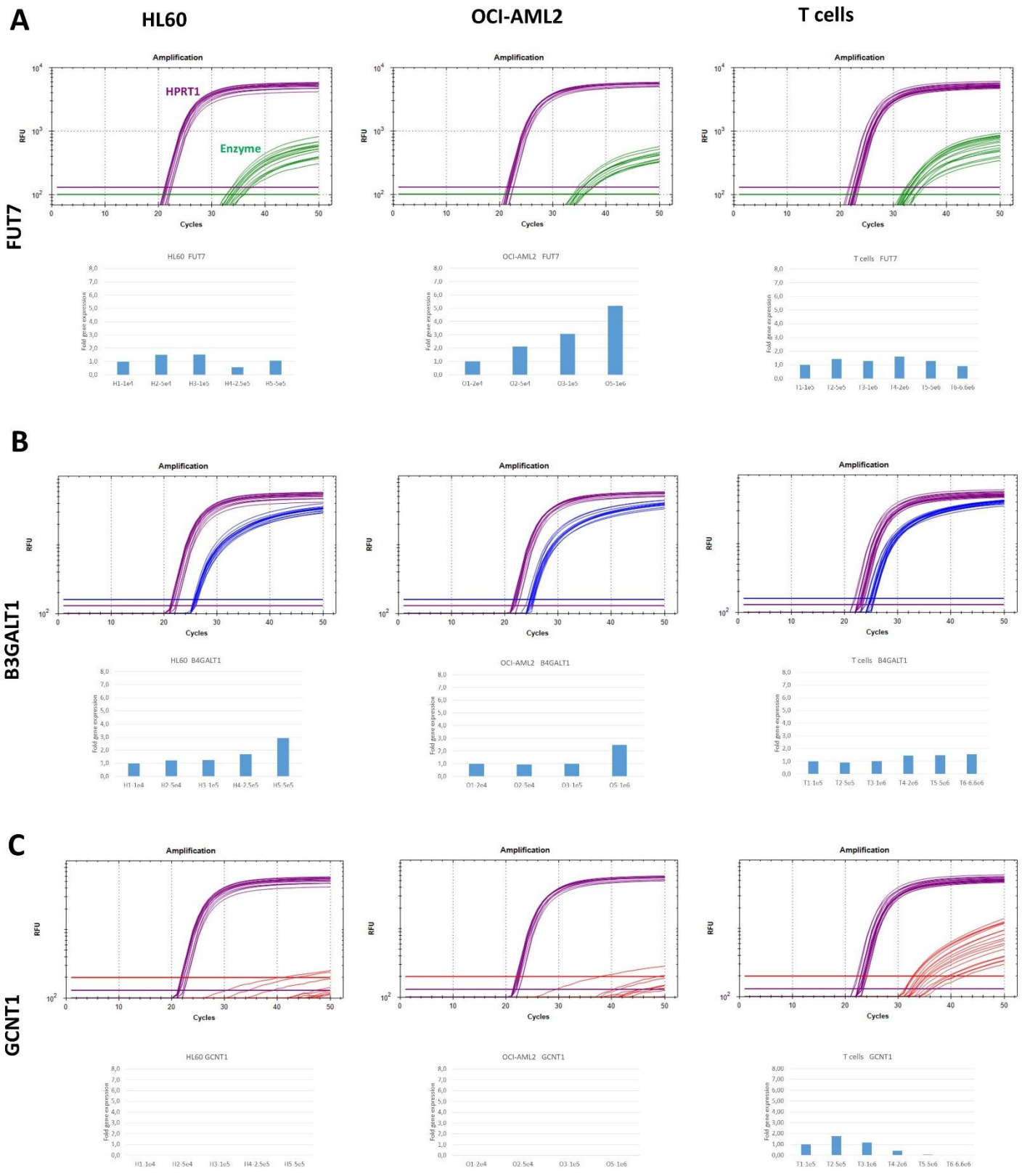
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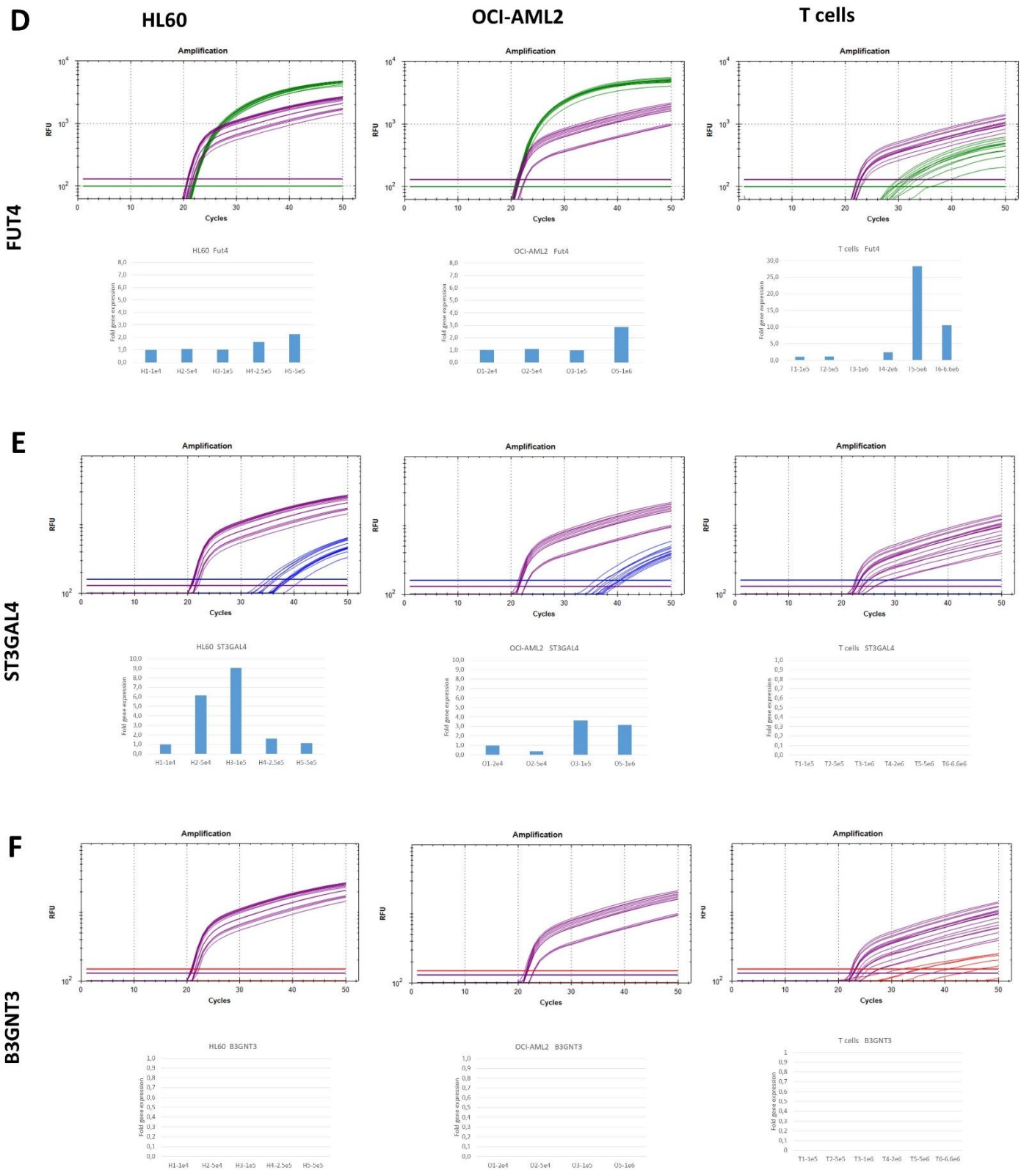
Supplementary figure S12. Investigation of CLA variability under three hypotheses: stress response, FBS concentration and cell density. ***A)*** *Schematic of experimental conditions testing the impact of stress, FBS or cell density on CLA expression in HL60 cells. Created with BioRender.com.* ***B)*** *CLA expression and cell*

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concentration over time in HL60 cells under various stress conditions. Conditions marked in red indicate treatments that induced widespread apoptosis or cell death. ***C)*** *CLA expression over time in HL60 cells cultured with varying FBS concentrations (0-20%).* ***D)*** *CLA expression over time in HL60 cells seeded at 2.5 × 10⁵ cells/mL under different splitting conditions on days 4 and 7: non-split control, 1:2 split with fresh media, and split leftovers without fresh media. Split cells showed reduced CLA expression compared to non-split controls.* ***E)*** *CLA expression over time at different starting cell densities in T cells, HL60 and OCI-AML2 cell lines.*



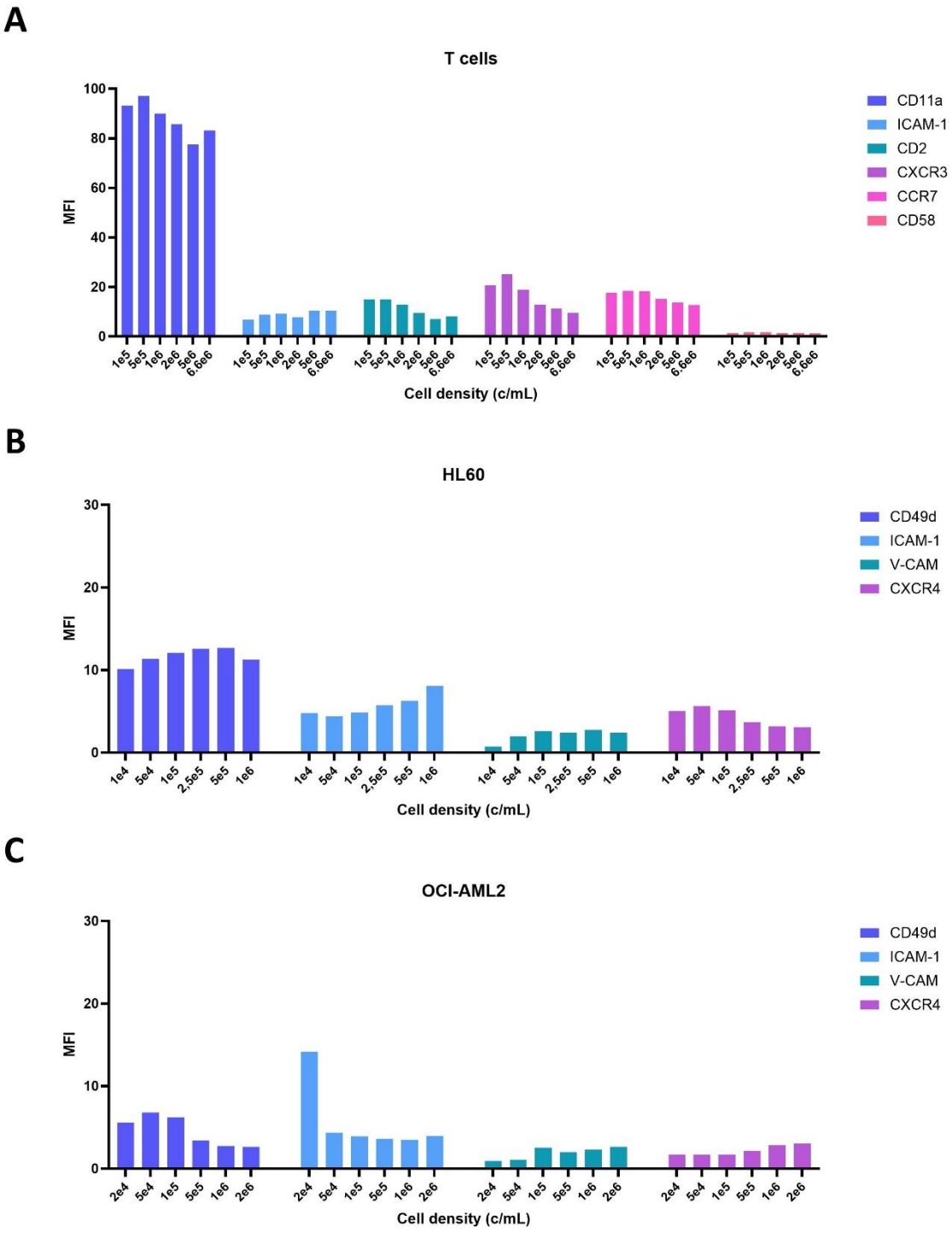
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Supplementary figure S13. RT-qPCR analysis of enzymes involved in CLA biosynthesis in multiple cell lines after 5 days of culture at different cell densities. *Cell lines analyzed: HL60, OCI-AML2 and peripheral T cells.* ***A-F)*** *Expression of key enzymes in CLA biosynthesis:* ***A)*** *α1,3-fucosyltransferase 7 (FUT7),* ***B)*** *β1,3-galactosyltransferase 1 (B3GALT1),* ***C)*** *β1,6-N-acetylglucos-aminyltransferase 1 (GCNT1),* ***D)*** *α1,3-fucosyltransferase 4 (FUT4),* ***E)*** *β-galactoside α2,3-sialyl-transferase 4 (ST3GAL4),* ***F)*** *β1,3-N-acetylglucos-aminyltransferase 3 (B3GNT3). For each enzyme and cell line, the amplification profile (cycle number versus relative fluorescence units [RFU]) is shown to illustrate PCR efficiency, with the HPRT1 housekeeping gene in purple. The bar plots below show the fold change in enzyme expression at the different*

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cell densities. Fold changes were calculated using the delta-delta Ct method and normalized to HPRT1 housekeeping gene.



Supplementary figure S14. Flow cytometry analysis of adhesion molecule regulation at varying cell densities. *Expression of adhesion molecules was assessed by mean fluorescence intensity (MFI) on T cells* ***A)****, HL60 cells* ***B)*** *or OCI-AML2 cells* ***C).*** *Cells were cultured for four days at densities ranging from 1 × 10⁴ to 6.6 × 10⁶ cells/mL, depending on the cell type, and analyzed by flow cytometry.*

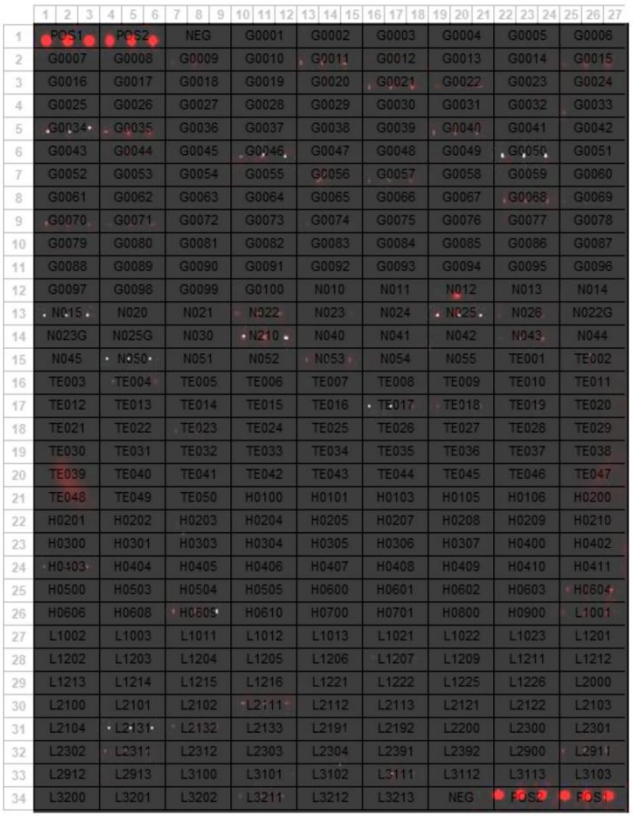
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A CLA REA (IgG)



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7	G0052			G0053			G0054			G0055			G0056			G0057			G0058			G0059			G0060	
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B CLA HECA 452 (IgM)



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C CD66c (IgG) – Negative control



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24	H0403			H0404			H0405			H0406			H0407			H0408			H0409			H0410			H0411	
25	H0500			H0503			H0504			H0505			H0600			H0601			H0602			H0603			H0604	
26	H0606			H0608			H0609			H0610			H0700			H0701			H0800			H0900			L1001	
27	L1002			L1003			L1011			L1012			L1013			L1021			L1022			L1023			L1201	
28	L1202			L1203			L1204			L1205			L1206			L1207			L1209			L1211			L1212	
29	L1213			L1214			L1215			L1216			L1221			L1222			L1225			L1226			L2000	
30	L2100			L2101			L2102			L2111			L2112			L2113			L2121			L2122			L2103	
31	L2104			L2131			L2132			L2133			L2191			L2192			L2200			L2300			L2301	
32	L2302			L2311			L2312			L2303			L2304			L2391			L2392			L2900			L2911	
33	L2912			L2913			L3100			L3101			L3102			L3111			L3112			L3113			L3103	
34	L3200			L3201			L3202			L3211			L3212			L3213			NEG			POS2			POS1	

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D CLA scFv-Fc VHVL

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27
1	POS1	POS1	POS1	POS2	POS2	POS2	NEG	NEG	NEG	G0001	G0001	G0001	G0002	G0002	G0002	G0003	G0003	G0003	G0004	G0004	G0004	G0005	G0005	G0005	G0006	G0006	G0006
2	G0007	G0007	G0007	G0008	G0008	G0008	G0009	G0009	G0009	G0010	G0010	G0010	G0011	G0011	G0011	G0012	G0012	G0012	G0013	G0013	G0013	G0014	G0014	G0014	G0015	G0015	G0015
3	G0016	G0016	G0016	G0017	G0017	G0017	G0018	G0018	G0018	G0019	G0019	G0019	G0020	G0020	G0020	G0021	G0021	G0021	G0022	G0022	G0022	G0023	G0023	G0023	G0024	G0024	G0024
4	G0025	G0025	G0025	G0026	G0026	G0026	G0027	G0027	G0027	G0028	G0028	G0028	G0029	G0029	G0029	G0030	G0030	G0030	G0031	G0031	G0031	G0032	G0032	G0032	G0033	G0033	G0033
5	G0034	G0034	G0034	G0035	G0035	G0035	G0036	G0036	G0036	G0037	G0037	G0037	G0038	G0038	G0038	G0039	G0039	G0039	G0040	G0040	G0040	G0041	G0041	G0041	G0042	G0042	G0042
6	G0043	G0043	G0043	G0044	G0044	G0044	G0045	G0045	G0045	G0046	G0046	G0046	G0047	G0047	G0047	G0048	G0048	G0048	G0049	G0049	G0049	G0050	G0050	G0050	G0051	G0051	G0051
7	G0052	G0052	G0052	G0053	G0053	G0053	G0054	G0054	G0054	G0055	G0055	G0055	G0056	G0056	G0056	G0057	G0057	G0057	G0058	G0058	G0058	G0059	G0059	G0059	G0060	G0060	G0060
8	G0061	G0061	G0061	G0062	G0062	G0062	G0063	G0063	G0063	G0064	G0064	G0064	G0065	G0065	G0065	G0066	G0066	G0066	G0067	G0067	G0067	G0068	G0068	G0068	G0069	G0069	G0069
9	G0070	G0070	G0070	G0071	G0071	G0071	G0072	G0072	G0072	G0073	G0073	G0073	G0074	G0074	G0074	G0075	G0075	G0075	G0076	G0076	G0076	G0077	G0077	G0077	G0078	G0078	G0078
10	G0079	G0079	G0079	G0080	G0080	G0080	G0081	G0081	G0081	G0082	G0082	G0082	G0083	G0083	G0083	G0084	G0084	G0084	G0085	G0085	G0085	G0086	G0086	G0086	G0087	G0087	G0087
11	G0088	G0088	G0088	G0089	G0089	G0089	G0090	G0090	G0090	G0091	G0091	G0091	G0092	G0092	G0092	G0093	G0093	G0093	G0094	G0094	G0094	G0095	G0095	G0095	G0096	G0096	G0096
12	G0097	G0097	G0097	G0098	G0098	G0098	G0099	G0099	G0099	G0100	G0100	G0100	N010	N010	N010	N011	N011	N011	N012	N012	N012	N013	N013	N013	N014	N014	N014
13	N015	N015	N015	N020	N020	N020	N021	N021	N021	N022	N022	N022	N023	N023	N023	N024	N024	N024	N025	N025	N025	N026	N026	N026	N022G	N022G	N022G
14	N023G	N023G	N023G	N025G	N025G	N025G	N030	N030	N030	N210	N210	N210	N040	N040	N040	N041	N041	N041	N042	N042	N042	N043	N043	N043	N044	N044	N044
15	N045	N045	N045	N050	N050	N050	N051	N051	N051	N052	N052	N052	N053	N053	N053	N054	N054	N054	N055	N055	N055	TE001	TE001	TE001	TE002	TE002	TE002
16	TE003	TE003	TE003	TE004	TE004	TE004	TE005	TE005	TE005	TE006	TE006	TE006	TE007	TE007	TE007	TE008	TE008	TE008	TE009	TE009	TE009	TE010	TE010	TE010	TE011	TE011	TE011
17	TE012	TE012	TE012	TE013	TE013	TE013	TE014	TE014	TE014	TE015	TE015	TE015	TE016	TE016	TE016	TE017	TE017	TE017	TE018	TE018	TE018	TE019	TE019	TE019	TE020	TE020	TE020
18	TE021	TE021	TE021	TE022	TE022	TE022	TE023	TE023	TE023	TE024	TE024	TE024	TE025	TE025	TE025	TE026	TE026	TE026	TE027	TE027	TE027	TE028	TE028	TE028	TE029	TE029	TE029
19	TE030	TE030	TE030	TE031	TE031	TE031	TE032	TE032	TE032	TE033	TE033	TE033	TE034	TE034	TE034	TE035	TE035	TE035	TE036	TE036	TE036	TE037	TE037	TE037	TE038	TE038	TE038
20	TE039	TE039	TE039	TE040	TE040	TE040	TE041	TE041	TE041	TE042	TE042	TE042	TE043	TE043	TE043	TE044	TE044	TE044	TE045	TE045	TE045	TE046	TE046	TE046	TE047	TE047	TE047
21	TE048	TE048	TE048	TE049	TE049	TE049	TE050	TE050	TE050	H0100	H0100	H0100	H0101	H0101	H0101	H0103	H0103	H0103	H0105	H0105	H0105	H0106	H0106	H0106	H0200	H0200	H0200
22	H0201	H0201	H0201	H0202	H0202	H0202	H0203	H0203	H0203	H0204	H0204	H0204	H0205	H0205	H0205	H0207	H0207	H0207	H0208	H0208	H0208	H0209	H0209	H0209	H0210	H0210	H0210
23	H0300	H0300	H0300	H0301	H0301	H0301	H0303	H0303	H0303	H0304	H0304	H0304	H0305	H0305	H0305	H0306	H0306	H0306	H0307	H0307	H0307	H0400	H0400	H0400	H0402	H0402	H0402
24	H0403	H0403	H0403	H0404	H0404	H0404	H0405	H0405	H0405	H0406	H0406	H0406	H0407	H0407	H0407	H0408	H0408	H0408	H0409	H0409	H0409	H0410	H0410	H0410	H0411	H0411	H0411
25	H0500	H0500	H0500	H0503	H0503	H0503	H0504	H0504	H0504	H0505	H0505	H0505	H0600	H0600	H0600	H0601	H0601	H0601	H0602	H0602	H0602	H0603	H0603	H0603	H0604	H0604	H0604
26	H0606	H0606	H0606	H0608	H0608	H0608	H0609	H0609	H0609	H0610	H0610	H0610	H0700	H0700	H0700	H0701	H0701	H0701	H0800	H0800	H0800	H0900	H0900	H0900	L1001	L1001	L1001
27	L1002	L1002	L1002	L1003	L1003	L1003	L1011	L1011	L1011	L1012	L1012	L1012	L1013	L1013	L1013	L1021	L1021	L1021	L1022	L1022	L1022	L1023	L1023	L1023	L1201	L1201	L1201
28	L1202	L1202	L1202	L1203	L1203	L1203	L1204	L1204	L1204	L1205	L1205	L1205	L1206	L1206	L1206	L1207	L1207	L1207	L1209	L1209	L1209	L1211	L1211	L1211	L1212	L1212	L1212
29	L1213	L1213	L1213	L1214	L1214	L1214	L1215	L1215	L1215	L1216	L1216	L1216	L1221	L1221	L1221	L1222	L1222	L1222	L1225	L1225	L1225	L1226	L1226	L1226	L2000	L2000	L2000
30	L2100	L2100	L2100	L2101	L2101	L2101	L2102	L2102	L2102	L2111	L2111	L2111	L2112	L2112	L2112	L2113	L2113	L2113	L2121	L2121	L2121	L2122	L2122	L2122	L2103	L2103	L2103
31	L2104	L2104	L2104	L2131	L2131	L2131	L2132	L2132	L2132	L2133	L2133	L2133	L2191	L2191	L2191	L2192	L2192	L2192	L2200	L2200	L2200	L2300	L2300	L2300	L2301	L2301	L2301
32	L2302	L2302	L2302	L2311	L2311	L2311	L2312	L2312	L2312	L2303	L2303	L2303	L2304	L2304	L2304	L2391	L2391	L2391	L2392	L2392	L2392	L2900	L2900	L2900	L2911	L2911	L2911
33	L2912	L2912	L2912	L2913	L2913	L2913	L3100	L3100	L3100	L3101	L3101	L3101	L3102	L3102	L3102	L3111	L3111	L3111	L3112	L3112	L3112	L3113	L3113	L3113	L3103	L3103	L3103
34	L3200	L3200	L3200	L3201	L3201	L3201	L3202	L3202	L3202	L3211	L3211	L3211	L3212	L3212	L3212	L3213	L3213	L3213	NEG	NEG	NEG	POS2	POS2	POS2	POS1	POS1	POS1

E CLA scFv-Fc VLVH

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27
1	POS1	POS2	NEG	G0001	G0002	G0003	G0004	G0005	G0006	G0007	G0008	G0009	G0010	G0011	G0012	G0013	G0014	G0015	G0016	G0017	G0018	G0019	G0020	G0021	G0022	G0023	G0024
2	G0025	G0026	G0027	G0028	G0029	G0030	G0031	G0032	G0033	G0034	G0035	G0036	G0037	G0038	G0039	G0040	G0041	G0042	G0043	G0044	G0045	G0046	G0047	G0048	G0049	G0050	G0051
3	G0052	G0053	G0054	G0055	G0056	G0057	G0058	G0059	G0060	G0061	G0062	G0063	G0064	G0065	G0066	G0067	G0068	G0069	G0070	G0071	G0072	G0073	G0074	G0075	G0076	G0077	G0078
4	G0079	G0080	G0081	G0082	G0083	G0084	G0085	G0086	G0087	G0088	G0089	G0090	G0091	G0092	G0093	G0094	G0095	G0096	G0097	G0098	G0099	G0100	N010	N011	N012	N013	N014
5	N015	N020	N021	N022	N023	N024	N025	N026	N022G	N023G	N025G	N030	N210	N040	N041	N042	N043	N044	N045	N050	N051	N052	N053	N054	N055	TE001	TE002
6	TE003	TE004	TE005	TE006	TE007	TE008	TE009	TE010	TE011	TE012	TE013	TE014	TE015	TE016	TE017	TE018	TE019	TE020	TE021	TE022	TE023	TE024	TE025	TE026	TE027	TE028	TE029
7	TE030	TE031	TE032	TE033	TE034	TE035	TE036	TE037	TE038	TE039	TE040	TE041	TE042	TE043	TE044	TE045	TE046	TE047	TE048	TE049	TE050	H0100	H0101	H0103	H0105	H0106	H0200
8	H0201	H0202	H0203	H0204	H0205	H0207	H0208	H0209	H0210	H0300	H0301	H0303	H0304	H0305	H0306	H0307	H0400	H0402	H0403	H0404	H0405	H0406	H0407	H0408	H0409	H0410	H0411
9	H0500	H0503	H0504	H0505	H0600	H0601	H0602	H0603	H0604	H0606	H0608	H0609	H0610	H0700	H0701	H0800	H0900	L1001	L1002	L1003	L1011	L1012	L1013	L1021	L1022	L1023	L1201
10	L1202	L1203	L1204	L1205	L1206	L1207	L1209	L1211	L1212	L1213	L1214	L1215	L1216	L1221	L1222	L1225	L1226	L2000	L2100	L2101	L2102	L2111	L2112	L2113	L2121	L2122	L2103
11	L2104	L2131	L2132	L2133	L2191	L2192	L2200	L2300	L2301	L2302	L2311	L2312	L2303	L2304	L2391	L2392	L2900	L2911	L2912	L2913	L3100	L3101	L3102	L3111	L3112	L3113	L3103
12	L3200	L3201	L3202	L3211	L3212	L3213	NEG	POS2	POS1																		

F

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27
1	POS1	POS1	POS1	POS2	POS2	POS2	NEG	NEG	NEG	G0001	G0001	G0001	G0002	G0002	G0002	G0003	G0003	G0003	G0004	G0004	G0004	G0005	G0005	G0005	G0006	G0006	G0006
2	G0007	G0007	G0007	G0008	G0008	G0008	G0009	G0009	G0009	G0010	G0010	G0010	G0011	G0011	G0011	G0012	G0012	G0012	G0013	G0013	G0013	G0014	G0014	G0014	G0015	G0015	G0015
3	G0016	G0016	G0016	G0017	G0017	G0017	G0018	G0018	G0018	G0019	G0019	G0019	G0020	G0020	G0020	G0021	G0021	G0021	G0022	G0022	G0022	G0023	G0023	G0023	G0024	G0024	G0024
4	G0025	G0025	G0025	G0026	G0026	G0026	G0027	G0027	G0027	G0028	G0028	G0028	G0029	G0029	G0029	G0030	G0030	G0030	G0031	G0031	G0031	G0032	G0032	G0032	G0033	G0033	G0033
5	G0034	G0034	G0034	G0035	G0035	G0035	G0036	G0036	G0036	G0037	G0037	G0037	G0038	G0038	G0038	G0039	G0039	G0039	G0040	G0040	G0040	G0041	G0041	G0041	G0042	G0042	G0042
6	G0043	G0043	G0043	G0044	G0044	G0044	G0045	G0045	G0045	G0046	G0046	G0046	G0047	G0047	G0047	G0048	G0048	G0048	G0049	G0049	G0049	G0050	G0050	G0050	G0051	G0051	G0051
7	G0052	G0052	G0052	G0053	G0053	G0053	G0054	G0054	G0054	G0055	G0055	G0055	G0056	G0056	G0056	G0057	G0057	G0057	G0058	G0058	G0058	G0059	G0059	G0059	G0060	G0060	G0060
8	G0061	G0061	G0061	G0062	G0062	G0062	G0063	G0063	G0063	G0064	G0064	G0064	G0065	G0065	G0065	G0066	G0066	G0066	G0067	G0067	G0067	G0068	G0068	G0068	G0069	G0069	G0069
9	G0070	G0070	G0070	G0071	G0071	G0071	G0072	G0072	G0072	G0073	G0073	G0073	G0074	G0074	G0074	G0075	G0075	G0075	G0076	G0076	G0076	G0077	G0077	G0077	G0078	G0078	G0078
10	G0079	G0079	G0079	G0080	G0080	G0080	G0081	G0081	G0081	G0082	G0082	G0082	G0083	G0083	G0083	G0084	G0084	G0084	G0085	G0085	G0085	G0086	G0086	G0086	G0087	G0087	G0087
11	G0088	G0088	G0088	G0089	G0089	G0089	G0090	G0090	G0090	G0091	G0091	G0091	G0092	G0092	G0092	G0093	G0093	G0093	G0094	G0094	G0094	G0095	G0095	G0095	G0096	G0096	G0096
12	G0097	G0097	G0097	G0098	G0098	G0098	G0099	G0099	G0099	G0100	G0100	G0100	N010	N010	N010	N011	N011	N011	N012	N012	N012	N013	N013	N013	N014	N014	N014
13	N015	N015	N015	N020	N020	N020	N021	N021	N021	N022	N022	N022	N023	N023	N023	N024	N024	N024	N025	N025	N025	N026	N026	N026	N022G	N022G	N022G
14	N023G	N023G	N023G	N025G	N025G	N025G	N030	N030	N030	N210	N210	N210	N040	N040	N040	N041	N041	N041	N042	N042	N042	N043	N043	N043	N044	N044	N044
15	N045	N045	N045	N050	N050	N050	N051	N051	N051	N052	N052	N052	N053	N053	N053	N054	N054	N054	N055	N055	N055	TE001	TE001	TE001	TE002	TE002	TE002
16	TE003	TE003	TE003	TE004	TE004	TE004	TE005	TE005	TE005	TE006	TE006	TE006	TE007	TE007	TE007	TE008	TE008	TE008	TE009	TE009	TE009	TE010	TE010	TE010	TE011	TE011	TE011
17	TE012	TE012	TE012	TE013	TE013	TE013	TE014	TE014	TE014	TE015	TE015	TE015	TE016	TE016	TE016	TE017	TE017	TE017	TE018	TE018	TE018	TE019	TE019	TE019	TE020	TE020	TE020
18	TE021	TE021	TE021	TE022	TE022	TE022	TE023	TE023	TE023	TE024	TE024	TE024	TE025	TE025	TE025	TE026	TE026	TE026	TE027	TE027	TE027	TE028	TE028	TE028	TE029	TE029	TE029
19	TE030	TE030	TE030	TE031	TE031	TE031	TE032	TE032	TE032	TE033	TE033	TE033	TE034	TE034	TE034	TE035	TE035	TE035	TE036	TE036	TE036	TE037	TE037	TE037	TE038	TE038	TE038
20	TE039	TE039	TE039	TE040	TE040	TE040	TE041	TE041	TE041	TE042	TE042	TE042	TE043	TE043	TE043	TE044	TE044	TE044	TE045	TE045	TE045	TE046	TE046	TE046	TE047	TE047	TE047
21	TE048	TE048	TE048	TE049	TE049	TE049	TE050	TE050	TE050	H0100	H0100	H0100	H0101	H0101	H0101	H0103	H0103	H0103	H0105	H0105	H0105	H0106	H0106	H0106	H0200	H0200	H0200
22	H0201	H0201	H0201	H0202	H0202	H0202	H0203	H0203	H0203	H0204	H0204	H0204	H0205	H0205	H0205	H0207	H0207	H0207	H0208	H0208	H0208	H0209	H0209	H0209	H0210	H0210	H0210
23	H0300	H0300	H0300	H0301	H0301	H0301	H0303	H0303	H0303	H0304	H0304	H0304	H0305	H0305	H0305	H0306	H0306	H0306	H0307	H0307	H0307	H0400	H0400	H0400	H0402	H0402	H0402
24	H0403	H0403	H0403	H0404	H0404	H0404	H0405	H0405	H0405	H0406	H0406	H0406	H0407	H0407	H0407	H0408	H0408	H0408	H0409	H0409	H0409	H0410	H0410	H0410	H0411	H0411	H0411
25	H0500	H0500	H0500	H0503	H0503	H0503	H0504	H0504	H0504	H0505	H0505	H0505	H0600	H0600	H0600	H0601	H0601	H0601	H0602	H0602	H0602	H0603	H0603	H0603	H0604	H0604	H0604
26	H0606	H0606	H0606	H0608	H0608	H0608	H0609	H0609	H0609	H0610	H0610	H0610	H0700	H0700	H0700	H0701	H0701	H0701	H0800	H0800	H0800	H0900	H0900	H0900	L1001	L1001	L1001
27	L1002	L1002	L1002	L1003	L1003	L1003	L1011	L1011	L1011	L1012	L1012	L1012	L1013	L1013	L1013	L1021	L1021	L1021	L1022	L1022	L1022	L1023	L1023	L1023	L1201	L1201	L1201
28	L1202	L1202	L1202	L1203	L1203	L1203	L1204	L1204	L1204	L1205	L1205	L1205	L1206	L1206	L1206	L1207	L1207	L1207	L1209	L1209	L1209	L1211	L1211	L1211	L1212	L1212	L1212
29	L1213	L1213	L1213	L1214	L1214	L1214	L1215	L1215	L1215	L1216	L1216	L1216	L1221	L1221	L1221	L1222	L1222	L1222	L1225	L1225	L1225	L1226	L1226	L1226	L2000	L2000	L2000
30	L2100	L2100	L2100	L2101	L2101	L2101	L2102	L2102	L2102	L2111	L2111	L2111	L2112	L2112	L2112	L2113	L2113	L2113	L2121	L2121	L2121	L2122	L2122	L2122	L2103	L2103	L2103
31	L2104	L2104	L2104	L2131	L2131	L2131	L2132	L2132	L2132	L2133	L2133	L2133	L2191	L2191	L2191	L2192	L2192	L2192	L2200	L2200	L2200	L2300	L2300	L2300	L2301	L2301	L2301
32	L2302	L2302	L2302	L2311	L2311	L2311	L2312	L2312	L2312	L2303	L2303	L2303	L2304	L2304	L2304	L2391	L2391	L2391	L2392	L2392	L2392	L2900	L2900	L2900	L2911	L2911	L2911
33	L2912	L2912	L2912	L2913	L2913	L2913	L3100	L3100	L3100	L3101	L3101	L3101	L3102	L3102	L3102	L3111	L3111	L3111	L3112	L3112	L3112	L3113	L3113	L3113	L3103	L3103	L3103
34	L3200	L3200	L3200	L3201	L3201	L3201	L3202	L3202	L3202	L3211	L3211	L3211	L3212	L3212	L3212	L3213	L3213	L3213	NEG	NEG	NEG	POS2	POS2	POS2	POS1	POS1	POS1

Supplementary figure S15. 300-glycan array scan images showing luminescent binding signals overlaid with the plate layout. *The assay was performed using* ***A)*** *CLA REA,* ***B)*** *CLA HECA-452,* ***C)*** *CD66c (background control),* ***D)*** *CLA scFv-Fc VHVL, and* ***E)*** *CLA scFv-Fc VLVH.* ***F)*** *Corresponding plate layout. Each sample was tested in triplicates. Raw and processed data, as well as full glycan names, are provided in the supplementary materials.*